
Multimodal AI Detection In Two Words

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Abstract

AI-generated text, images, audio, and video are now ubiquitous and difficult for humans to reliably identify. Scalable, automated detection is therefore a practical necessity. Existing detectors are fragmented along modality lines and not faithfully explainable. We propose a unified method that addresses both issues by extending *neologism learning* from text generation steering to multimodal classification. We add two tokens, <REAL> and <AIGEN>, to the vocabulary of an otherwise-frozen multimodal LLM (MLLM) and train only their embedding rows (*2d parameters, where d is the backbone’s token-embedding dimension; fewer than 0.001% of the backbone*) so that the model’s next-token distribution under a paired prompt encodes the real-vs-AI-generated decision. Because all modalities are projected into a shared input embedding space, the same token pair applies across text, image, audio, and video without per-modality heads, and a pair trained on a subset of modalities *transfers above chance to several, though not all, held-out ones*. The trained tokens further admit free-form natural-language descriptions of what <AIGEN> has come to mean, decoded from the same frozen backbone and never supervised during training; we validate their causal faithfulness via plug-in evaluation. Across standard text, audio, image, and video benchmarks, our method matches or exceeds dedicated, modality-specific detectors and *is on par with the strongest MLLM-based explainable detector out of distribution*, despite training orders of magnitude fewer parameters and using no human-annotated explanations.

1 Introduction

Generative AI for text, images, video, and audio now produces synthetic content that is both ubiquitous and difficult to identify [Liu et al., 2024b, Cao, 2025, Livernoche et al., 2025, Dolezal et al., 2026], enabling misinformation [Chandra et al., 2025, Livernoche et al., 2025], identity fraud [Khalid et al., 2021], non-consensual imagery, fabricated evidence, and academic dishonesty [Cao, 2025]. Humans are already largely incapable of reliably distinguishing real from synthetic content [Cooke et al., 2024, Barrington et al., 2025], and the gap may widen as generation improves.

Scalable, automated AI detection has thus become a practical necessity for journalists, fact-checkers, educators, courts, and content moderators. Critically, users need to know not merely *whether* content is AI-generated but also *why*. Saliency explanations [Selvaraju et al., 2017] highlight regions but rarely identify what is wrong with them, with reliability concerns of their own [Tsigos et al., 2024]. The community has thus moved toward natural-language explanations [Zhang et al., 2024, Li et al., 2025b, Liu et al., 2024a, Xu et al., 2024, Huang et al., 2025b, Hondru et al., 2026, Tariq et al., 2025, Yang et al., 2025], but these rely on human-annotated rationales: the explanations may not reflect what the detector actually learned, and the supervision becomes increasingly unreliable as human ability to spot AI content degrades.

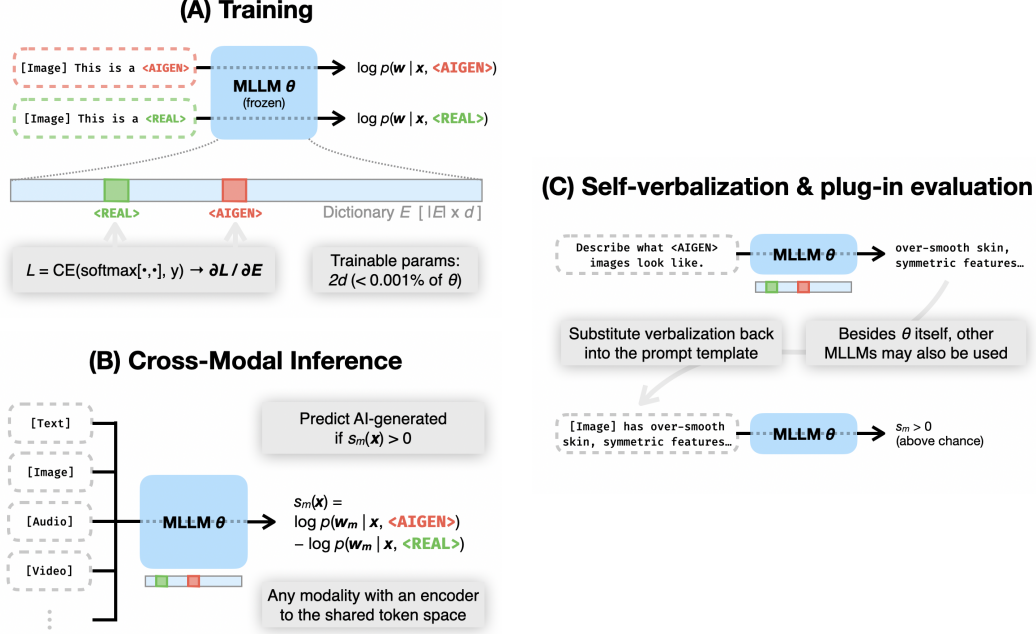


Figure 1: **Method overview.** (A) **Training.** We add tokens $\langle \text{REAL} \rangle$ and $\langle \text{AIGEN} \rangle$ to a frozen MLLM θ and pair each with input x in contrastive prompts; only the two corresponding rows of E get updated based on log-prob cross-entropy of a fixed continuation w . (B) **Cross-modal inference.** These two rows define a detector $s_m(x) = \log p(w_m | x, \langle \text{AIGEN} \rangle) - \log p(w_m | x, \langle \text{REAL} \rangle)$, predicting AI-generated when $s_m(x) > 0$; the same pair applies to any modality the backbone accepts at input. (C) **Self-verbalization & plug-in evaluation.** We prompt θ to describe each token in natural language (top); plug-in evaluation substitutes the description back into the prompt and rescores (bottom), with above-chance s_m indicating faithfulness.

AI detection is mostly fragmented along modality lines. Most methods operate on a single modality, with separate datasets and pipelines for images [Ojha et al., 2023, Khan and Dang-Nguyen, 2024, Cozzolino et al., 2024, Pellegrini et al., 2025, Chandra et al., 2025], video [Bohacek and Farid, 2024a], audio, or text. Existing multimodal methods are almost exclusively *audio-visual*, based on inconsistencies between audio and face video [Khalid et al., 2021, Ilyas et al., 2023, Yang et al., 2023, Wang et al., 2025a, Zhong et al., 2024, Bohacek and Farid, 2024b, 2022, Barrington et al., 2024, Agarwal et al., 2020].

Recent work has begun adapting vision-language models (VLMs) and multimodal LLMs (MLLMs) to AI detection. Zero-shot prompting of frontier MLLMs underperforms task-specific detectors [Jia et al., 2024, Shi et al., 2025, Ojha et al., 2023, Khan and Dang-Nguyen, 2024, Cozzolino et al., 2024]. MLLM-based explainable detectors [Zhang et al., 2024, Li et al., 2025b, Liu et al., 2024a, Xu et al., 2024, Huang et al., 2025b, Hondru et al., 2026] are uniformly trained on human-annotated rationales, so their explanations are supervised rather than derived from what the detector has learned. The closest cross-modality analog, Huang et al. [2025a], applies a shared encoder [Girdhar et al., 2023, Zhu et al., 2023a] to text, image, and voice but retains per-modality heads and offers no explanations; Wen et al. [2025] covers image and video only; Li et al. [2025a] covers only watermarking. To our knowledge, no prior work unifies cross-modality detection in a single shared decision representation or derives natural-language explanations without explanation supervision.

This work. We address both fragmentation and supervision overhead by extending *neologism learning* [Hewitt et al., 2025a,b, Park et al., 2025]—a parameter-efficient technique that introduces new word embeddings into a frozen LM—from LM steering to multimodal classification. As illustrated in Fig. 1, we add two tokens, $\langle \text{REAL} \rangle$ and $\langle \text{AIGEN} \rangle$, to a MLLM’s vocabulary and train only

their embedding rows, so that the model assigns higher probability to a designated continuation depending on whether the context is real or AI-generated. Our contributions are as follows:

- **A unified and efficient method for AI-generated content detection across modalities (Sec. 3).** We train a single pair of token embeddings in a frozen MLLM ($\sim 2d$ parameters, $< 0.001\%$ of the backbone) to discriminate AI-generated from human-produced content, applying the same representation regardless of input modality and eliminating modality-specific classifier heads.
- **Competitive detection (Sec. 4.1) and partial cross-modal transfer (Sec. 4.3) through a shared embedding.** On standard benchmarks, our approach is competitive with dedicated embedding- and MLLM-based detectors despite training orders of magnitude fewer parameters. Because the $\langle \text{AIGEN} \rangle$ embedding lives in the LLM’s modality-aligned latent, [a pair trained on a subset of modalities can be evaluated on any modality the backbone accepts, and transfers above chance to several—though not all—held-out modalities at test time.](#)
- **Faithful natural-language explanations without explanation supervision (Sec. 4.7).** We elicit free-form self-verbalizations of $\langle \text{AIGEN} \rangle$ and $\langle \text{REAL} \rangle$; unlike prior explainable detectors, our model is never shown a textual rationale during training, so the descriptions reflect features the embedding has actually learned. We validate them via *plug-in evaluation* [Hewitt et al., 2025b], replacing the trained token with its verbalization and measuring detection performance.

2 Preliminaries: Neologism Learning and Self-Verbalization

Our method extends neologism learning [Hewitt et al., 2025a,b] for multimodal classification. We briefly review the framework and self-verbalization, both exploited by our method.

Notation. Let $\mathcal{M}_\theta$ denote a pretrained, possibly multimodal LM with parameters θ , vocabulary $\mathcal{V}$, and input embedding matrix $\mathbf{E} \in \mathbb{R}^{d \times |\mathcal{V}|}$, whose row $\mathbf{E}_v \in \mathbb{R}^d$ embeds token v . For an input sequence $\mathbf{x}_{<t}$, the model produces a next-token distribution $p_\theta(\cdot \mid \mathbf{x}_{<t}) \in \Delta^{|\mathcal{V}|}$, with the output projection typically tied to $\mathbf{E}^\top$. When $\mathcal{M}_\theta$ supports non-text modalities, modality-specific encoders project image patches, audio frames, and video frames into the same d -dimensional latent as $\mathbf{E}$ [Girdhar et al., 2023, Zhu et al., 2023a, Grattafiori et al., 2024, Google DeepMind, 2026]; the backbone then sees a heterogeneous sequence of token vectors in a shared space.

Neologism learning. Hewitt et al. [2025a,b] introduce neologism learning as a parameter-efficient method for steering an otherwise-frozen LM toward a desired concept. One adds a token $c \notin \mathcal{V}$ to the vocabulary, $\mathcal{V}' = \mathcal{V} \cup \{c\}$, with a corresponding row $\mathbf{E}_c \in \mathbb{R}^d$ in the embedding matrix; with θ frozen, only $\mathbf{E}_c \in \mathbb{R}^d$ is updated via gradient descent on a behavioral loss over input-output examples of the target concept. The trained $\mathbf{E}_c$ thus encodes the entire concept as a single direction in the LM’s input embedding space, with a footprint of only d scalars [Park et al., 2025].

Self-verbalization. Because θ is unchanged, the LM can produce descriptions of what c has come to mean [Hewitt et al., 2025b]. One places c inside a question such as “*List five synonyms for c* ” or “*Describe what c responses are characterized by*” and decodes a completion; since $\mathbf{E}_c$ now sits among existing vocabulary directions, the decoder treats it as any other word. Self-verbalizations are not guaranteed to be faithful, so Hewitt et al. [2025b] propose *plug-in evaluation*: substituting the verbalization back into the original behavioral prompt, a verbalization that steers the model similarly to c itself is deemed faithful.

2.1 Relation to Rationale-Supervised Explainable Detectors

The line of MLLM-based explainable detectors closest to our setting [Zhang et al., 2024, Li et al., 2025b, Liu et al., 2024a, Xu et al., 2024, Huang et al., 2025b, Hondru et al., 2026] also emits a natural-language account of its decision, but obtains it in a fundamentally different way. We make the distinction explicit along four axes.

Explanation source. Rationale-supervised detectors are trained on human-written explanations and therefore *imitate* an annotator; our descriptions are decoded from $\mathbf{E}_{\langle \text{AIGEN} \rangle}$ through an otherwise-unchanged backbone and are never part of the training signal. *Annotation cost.* They require a per-example rationale in addition to the label; we require only the binary label already present in every

detection benchmark. *Failure mode as generators improve.* Their supervision inherits the ceiling of human ability to articulate what makes content synthetic, which is already low and declining [Cooke et al., 2024, Barrington et al., 2025]; our signal is learned from labels alone, so it is not bounded by what an annotator could name. *Verifiability.* A rationale-supervised explanation is optimized to be plausible, which makes plausibility uninformative about faithfulness. Ours is optimized only to be discriminative, so faithfulness becomes an empirical question that can be tested—which is what plug-in evaluation (Sec. 3.2) does.

3 Method

We extend neologism learning from generation steering to multimodal binary classification. We introduce two tokens, <REAL> and <AIGEN>, and train their embeddings so that the LM’s next-token distribution under a paired prompt template encodes the real-vs-AI-generated decision. Throughout, θ is frozen and only $(\mathbf{E}_{\text{<REAL>}}, \mathbf{E}_{\text{<AIGEN>}}) \in \mathbb{R}^{2d}$ is trainable. The pipeline is illustrated in Fig. 1.

3.1 Detection via Token-Pair Log-Probability Comparison

Let $\mathcal{X}_m$ denote the input space for modality $m \in \mathcal{M} := \{\text{text, image, audio, video}\}$, and let $y \in \{0, 1\}$ encode the binary label (0 = human-produced, 1 = AI-generated). For each modality we define a *prompt template* $\Pi_m(\tau, \mathbf{x})$, placing a token τ alongside an input $\mathbf{x} \in \mathcal{X}_m$ inside a user turn:

$$\begin{aligned}\Pi_{\text{image}}(\tau, \mathbf{x}) &= \tau \langle \text{image}(\mathbf{x}) \rangle \text{ “This is a } \tau\text{”}, \\ \Pi_{\text{text}}(\tau, \mathbf{x}) &= \tau \mathbf{x} \text{ “This is a } \tau\text{”},\end{aligned}$$

with analogous constructions for audio and video. The template is instantiated twice per input—once with $\tau = \text{<REAL>}$ and once with $\tau = \text{<AIGEN>}$ —and each instance is passed through $\mathcal{M}_\theta$. We then evaluate the log-probability the model assigns to a fixed *continuation token* $w_m \in \mathcal{V}$ at the next-token position, where w_m names the modality in plain English (e.g., $w_{\text{image}} = \text{Image}$, $w_{\text{audio}} = \text{Audio}$). Pinning a concrete completion under both prompts isolates the discriminative signal to the difference between $\mathbf{E}_{\text{<REAL>}}$ and $\mathbf{E}_{\text{<AIGEN>}}$, rather than reflecting marginal preferences over any particular completion. We define the modality- m score:

$$s_m(\mathbf{x}) := \log p_\theta(w_m \mid \Pi_m(\text{<AIGEN>}, \mathbf{x})) - \log p_\theta(w_m \mid \Pi_m(\text{<REAL>}, \mathbf{x})), \quad (1)$$

and predict $\hat{y}(\mathbf{x}) = \mathbb{1}[s_m(\mathbf{x}) > 0]$. Because s_m is a real-valued score rather than a hard label, it supports threshold-based ROC analysis without retraining.

Training objective. Given a labeled set $\mathcal{D}_m = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, we treat the two log-probabilities from Eq. 1 as a 2-class logit vector and minimize cross-entropy:

$$\begin{aligned}z_i &= (\log p_\theta(w_m \mid \Pi_m(\text{<REAL>}, \mathbf{x}_i)), \log p_\theta(w_m \mid \Pi_m(\text{<AIGEN>}, \mathbf{x}_i))), \quad (2) \\ \mathcal{L}(\mathbf{E}_{\text{<REAL>}}, \mathbf{E}_{\text{<AIGEN>}}) &= -\frac{1}{N} \sum_{i=1}^N \log \text{softmax}(z_i)_{y_i}.\end{aligned} \quad (3)$$

All of θ except $(\mathbf{E}_{\text{<REAL>}}, \mathbf{E}_{\text{<AIGEN>}})$ is frozen, and gradients on $\mathbf{E}$ are elementwise-masked so updates flow only into those two rows. Both rows are warm-started from a neutral existing token $w_0 \in \mathcal{V}$ (modality-appropriate noun, e.g., “writing” for text; see App. A for the per-modality choices as resolved by each tokenizer), so that post-training divergence between them reflects the real-vs-AI distinction rather than incidental properties of the warm-start word.

Generalization across modalities. Eq. 1 requires only that (i) $\mathcal{M}_\theta$ accepts modality- m inputs into its prompt and (ii) a sensible continuation token $w_m \in \mathcal{V}$ exists. The trainable parameters $(\mathbf{E}_{\text{<REAL>}}, \mathbf{E}_{\text{<AIGEN>}})$ are not modality-specific by construction: they are vectors in the same d -dimensional input embedding space into which all modalities are projected. A single trained pair can therefore be evaluated against any modality the backbone supports, including modalities unseen at training time, by substituting only Π_m and w_m .

Joint multimodal training. When training data is available for a subset of modalities $\mathcal{M}_{\text{train}} \subseteq \mathcal{M}$, we train the embedding pair jointly across all of them. At each step a batch is drawn from one modality’s loader, rotating through $\mathcal{M}_{\text{train}}$ in round-robin order; the per-modality loss (Eq. 3) uses that modality’s continuation token w_m and is accumulated into the shared embeddings via gradient accumulation. Because the pair is modality-agnostic (Sec. 3.1), gradient signal from any one modality contributes to a common notion of real vs. AI-generated rather than to a per-modality classifier.

3.2 Self-Verbalization and Plug-in Evaluation

Once $(\mathbf{E}_{\langle \text{REAL} \rangle}, \mathbf{E}_{\langle \text{AIGEN} \rangle})$ are trained, $\mathcal{M}_\theta$ treats both tokens as ordinary vocabulary entries and can be queried about them. Since the backbone is otherwise unchanged, any description of the two tokens reflects only what the trained embeddings encode. We elicit two complementary forms of self-verbalization, both following the forced-start protocol of Hewitt et al. [2025b].

Global verbalization. A concept-level probe asks $\mathcal{M}_\theta$ what $\langle \text{AIGEN} \rangle$ and $\langle \text{REAL} \rangle$ mean with no datapoint in context, constraining the assistant turn to begin with a fixed prefix such as:

User: *The concept $\langle \text{AIGEN} \rangle$ is used to classify $\langle \text{modality} \rangle$. Describe what $\langle \text{AIGEN} \rangle$ $\langle \text{modality} \rangle$ look (resp. sound, read) like.*
Assistant (forced start): *Ok, $\langle \text{AIGEN} \rangle$ $\langle \text{modality} \rangle$ are characterized by ...*

The continuation is the model’s free-form articulation of the regularities its embedding has captured.

Instance verbalization. A datapoint-conditioned probe places a specific input $\mathbf{x} \in \mathcal{X}_m$ alongside the question, asking $\mathcal{M}_\theta$ to explain what *this particular $\mathbf{x}$* exhibits that makes $\langle \text{AIGEN} \rangle$ apply (or fail to). The same forced-start mechanism is used.

Plug-in evaluation. A verbalization is faithful if substituting it for the trained token reproduces the detection behavior. We replace $\tau \in \{\langle \text{REAL} \rangle, \langle \text{AIGEN} \rangle\}$ in Π_m with the natural-language description $\nu(\tau)$ produced by global verbalization, recompute $s_m(\mathbf{x})$ on a held-out test set under the substituted prompts, and compare against the trained-token score. The verbalization is faithful if it still yields detection well above chance—not necessarily matching the trained tokens, but indicating that the description carries genuine signal about the learned embedding.

3.3 What the Trained Pair Computes

The construction admits a compact characterization that explains why two embedding rows suffice. Write $\mathbf{h}_m(\tau, \mathbf{x}) \in \mathbb{R}^d$ for the residual-stream state at the continuation position under $\Pi_m(\tau, \mathbf{x})$. Because the output projection is tied to $\mathbf{E}^\top$, the logit assigned to w_m is $\mathbf{E}_{w_m}^\top \mathbf{h}_m(\tau, \mathbf{x})$, and Eq. 1 becomes

$$s_m(\mathbf{x}) = \mathbf{E}_{w_m}^\top (\mathbf{h}_m(\langle \text{AIGEN} \rangle, \mathbf{x}) - \mathbf{h}_m(\langle \text{REAL} \rangle, \mathbf{x})) - (\log Z_{\langle \text{AIGEN} \rangle} - \log Z_{\langle \text{REAL} \rangle}), \quad (4)$$

with $\log Z_\tau$ the log-partition term of the corresponding next-token distribution. The two prompts are identical except for the single token embedding placed at the τ positions, so to first order in the perturbation $\boldsymbol{\delta} := \mathbf{E}_{\langle \text{AIGEN} \rangle} - \mathbf{E}_{\langle \text{REAL} \rangle}$ the state difference is $\mathbf{J}_m(\mathbf{x}) \boldsymbol{\delta}$, where $\mathbf{J}_m(\mathbf{x})$ is the Jacobian of the residual state with respect to the injected embedding. Eq. 4 then reads

$$s_m(\mathbf{x}) \approx \langle \mathbf{J}_m(\mathbf{x})^\top \mathbf{E}_{w_m}, \boldsymbol{\delta} \rangle + \text{const}, \quad (5)$$

i.e. an inner product between an input-dependent feature vector computed by the frozen backbone and the single *difference direction* $\boldsymbol{\delta}$. Training therefore fits one direction, and detection reduces to a linear probe on a frozen representation—which is consistent with the single-direction findings of Park et al. [2025] and Hewitt et al. [2025a,b]. What distinguishes it from an ordinary probe is where the direction lives: because $\boldsymbol{\delta}$ is a difference of *vocabulary* rows and the LM head is tied to $\mathbf{E}$, the same readout can be decoded back into language (Sec. 3.2), and any modality whose encoder writes into the same latent supplies its own $\mathbf{J}_m$ without new parameters (Sec. 3.1). We stress that Eq. 5 characterizes the functional form of the method, not a generalization guarantee; the empirical scope of the shared direction is the subject of Sec. 4.3.

4 Results

We evaluate four claims of the method: (i) a single trained embedding pair matches dedicated, modality-specific detectors across text, audio, image, and video (Sec. 4.1); (ii) the pair generalizes both across datasets and across unseen generators within a dataset (Sec. 4.2); (iii) the pair transfers across [some, though not all](#), modality pairs, including to held-out modalities at test time (Sec. 4.3); and (iv) the trained tokens admit faithful, free-form natural-language descriptions that point at features actually used by the detector (Secs. 4.6–4.7). All detection metrics are reported as [the area under the receiver-operating-characteristic curve \(AUROC\)](#). Unless stated otherwise, results use Llama-3.1-8B-AI-Neo for unimodal evaluations and Gemma-4-E4B-AI-Neo for multimodal evaluations.

Throughout, *in-distribution* (ID) denotes the evaluation split of the dataset a model was trained on and *out-of-distribution* (OOD) a disjoint benchmark seen only at test time. Every number we report for our own models carries a percentile-bootstrap 95% confidence interval over test examples (10,000 resamples, fixed RNG); baseline entries are point estimates, since we do not have per-sample scores for released checkpoints. Evaluation sets are drawn from each source dataset’s own test split by a seeded, class-balanced procedure documented in App. A.5, and the resulting sample lists and per-sample scores are part of our release.

4.1 A two-row embedding edit matches modality-specific SOTA

We compare against state-of-the-art open-source detectors on standard benchmarks for each of the four modalities: RAID [Dugan et al., 2024] (text), AUDETER [Wang et al., 2025b] (audio), GenImage [Zhu et al., 2023b] (image), and DFE-2024 [Chandra et al., 2025] (video). Baselines were chosen as the strongest publicly available method per modality that reports its training mixture and only includes the training split of the benchmark: DivEye [Basani and Chen, 2025], Binoculars [Hans et al., 2024], TMR Oxidane [2025], and a custom linear classifier for text; AASIST [Jung et al., 2022] and RawNet2 [Tak et al., 2021], for audio; UnivFD [Ojha et al., 2023] and AIDE [Yan et al., 2024] for image; and SBI [Shiohara and Yamasaki, 2022] and CDFA [Lin et al., 2024] for video.

Tab. 1 (left) reports ID AUROC. Our models saturate on text and audio, matching respective baselines (TMR-Detector, within 0.02 AUROC for Llama-AI-Neo, and AASIST/RawNet2), and exceed all baselines on image ([+0.08 over AIDE](#)) and video ([+0.02 over SBI](#)), despite training only the two embedding rows on top of an otherwise frozen backbone. [The image and video rows are evaluated at 20× the sample size of the submitted version \(\$n = 2,000\$ for image; the complete DFE-2024 test split for video\)](#), which shrinks the video margin over SBI and CDFA from +0.07 to +0.02 while leaving the ordering intact. Training with three random seeds moves ID AUROC by at most 0.02 (App. A.5).

The recipe is not specific to one backbone. We repeated the Tab. 1 protocol unchanged on two further backbones, Gemma-3n-E4B-it and Qwen2.5-VL-7B-Instruct, the latter from a different vendor with a different, dynamic-resolution vision encoder. In-distribution performance reproduces on both, and on the generator-shift benchmark of Sec. 4.2 Gemma-4 and Qwen2.5-VL land within 0.01 AUROC of one another. The only backbone-specific change required was placing <REAL> and <AIGEN> in two reserved in-range vocabulary slots on Gemma-3n, whose vocabulary tail is routed to the vision and audio embedders. Full numbers are in App. C.1.

4.2 Detection holds across datasets and across unseen generators

Cross-dataset OOD. Each model is evaluated, without further adaptation, on a second benchmark covering the same modality but disjoint generators and content sources: Beemo [Artemova et al., 2025] (text), In-the-Wild [Müller et al., 2022] (audio), CIFAKE [Bird and Lotfi, 2024] (image), and DeeprtraceReward [Fu et al., 2025] (video). Our pair is the strongest method on image and video under the shift, and trails the strongest baseline by [0.05 AUROC on text and 0.09 on audio](#) (right column of Tab. 1). The dedicated baselines that match us in-distribution incur the largest shifts: DivEye drops −0.34 on text and AIDE −0.28 on image.

Table 1: **Detection Performance.** The models are tested on the in-distribution (ID) dataset (evaluation split) and held-out out-of-distribution (OOD) dataset; AUROC is reported. Our rows carry percentile-bootstrap 95% CIs over test examples; $\pm$ denotes the standard deviation over three training seeds where no bootstrap was run (App. A.5).

Method	ID AUROC	OOD AUROC
<i>Text</i> — RAID $\rightarrow$ Beemo		
DivEye	0.91	0.57
Binoculars	0.85	0.84
TMR-Detector	1.00	0.77
Llama-3.1-8B-Classifier	0.99	0.72
Gemma-4-E4B-Classifier	0.97	0.71
Llama-3.1-8B-AI-Neo	0.98 ± 0.01	0.79 ± 0.01
Gemma-4-E4B-AI-Neo	$0.96 [0.95, 0.96]$	$0.77 [0.76, 0.78]$
<i>Audio</i> — AUDETER $\rightarrow$ In-the-Wild		
AASIST	1.00	0.79
RawNet2	1.00	0.77
Gemma-4-E4B-AI-Neo	$1.00 [1.00, 1.00]$	$0.70 [0.68, 0.73]$
<i>Image</i> — GenImage $\rightarrow$ CIFAKE		
UnivFD	0.56	0.47
AIDE	0.82	0.54
Gemma-4-E4B-AI-Neo	$0.90 [0.89, 0.91]$	$0.61 [0.58, 0.63]$
<i>Video</i> — DFE-2024 $\rightarrow$ DeepttraceReward		
SBI	0.76	0.58
CDFA	0.75	0.60
Gemma-4-E4B-AI-Neo	$0.78 [0.75, 0.81]$	$0.87 [0.84, 0.90]$

Table 2: **Generator Generalization.** DivEye, Binoculars, and Llama-3.1-8B-AI-Neo (LL-AI-Neo) are tested for generator generalization on the task of AI-generated text detection. For each held-out generator g , models train on human samples plus the remaining $G-1$ generators and evaluate on human plus g ; AUROC is reported.

Held-out gen.	DivEye	Bino	LI-AI-Neo
MPT	0.90	0.59	0.94
Cohere	0.80	0.93	0.96
Mistral	0.85	0.71	0.96
Cohere Chat	0.84	0.95	0.98
GPT-3	0.91	0.97	0.98
GPT-2	0.96	0.82	0.99
GPT-4	0.82	0.94	1.00
Mistral Chat	0.88	0.93	1.00
MPT Chat	0.87	0.77	1.00
ChatGPT	0.89	0.97	1.00
Llama Chat	0.87	0.97	1.00
Mean	0.87	0.87	0.98
Worst-case	0.80	0.59	0.94

Separating generator shift from content-domain shift. The image row deserves a caveat that applies to every method in it. CIFAKE pairs 32×32 CIFAR-10 content against a single generator, whereas our ID training images are 224×224 and drawn from an ImageNet-like distribution, so the CIFAKE column measures resolution and content-domain shift at least as much as generator shift—a confound also reported by Grommelt et al. [2024] and Hong et al. [2024]. Per-sample scores make the mechanism visible: on CIFAKE *every* image scores positive (class means $+2.39$ real / $+2.61$ AI, against -1.25 / $+1.20$ in distribution), i.e. the learned direction partly entangles “AI-generated” with “unlike the training domain”. To isolate the generator axis we therefore add GenImage++ [Zhou et al., 2025], a test-only benchmark whose modern generators are prompted from the same content distribution as GenImage; we put both classes through one preprocessing pipeline so that resolution cannot act as a class cue. The same unchanged checkpoint reaches 0.92 [0.90, 0.93] on FLUX.1 and 0.89 [0.87, 0.90] on SD3, at $n = 2,000$ each, against 0.61 [0.58, 0.63] on CIFAKE. Under pure generator shift the pair therefore transfers substantially, and the CIFAKE number reflects the compounded shift rather than a failure to generalize across generators. Because the trainable footprint is $2d$ scalars, the residual gap is also cheap to close by adaptation rather than retraining: fitting only the two rows on 1,000 labeled pairs from the CIFAKE *training* split, leaving the test split untouched, raises CIFAKE to 0.97, and the same protocol on OpenFake [Livermoche et al., 2025] moves 0.64 to 0.97 (App. C.5).

Within-dataset leave-one-generator-out. We run a leave-one-generator-out protocol on RAID [Dugan et al., 2024], which contains human-written text and AI-generated text from $G = 11$ distinct generators, with results reported in Tab. 2. Our pair attains mean held-out AUROC 0.98 with worst-case fold 0.94, against 0.87 mean and 0.59–0.80 worst-case for the two strongest text baselines under the same protocol. This result suggests that our method captures regularities that survive across generator families.

Two observations qualify what this protocol shows. First, any two folds share roughly 90% of their training data, so the G learned tokens are near-identical (mean pairwise cosine 0.94 when each is instead trained on a single generator in isolation); the shared <AIGEN> embedding is picking up discourse-level regularities that hold *across* generator families, which is also why leave-one-generator-out barely degrades. Generator-specific fingerprints are better surfaced by the contrastive pairwise setup of Sec. 4.7. Second, difficulty is concentrated on the AI side and tracks generator recency: on the image analogue of this analysis, held-out AUROC ranges from 0.99 on BigGAN to 0.82 on Midjourney across three seeds, while real-image scores stay flat (App. C.6).

4.3 Cross-modality transfer is available between some pairs

We next ask whether the modality-agnostic property of $(\mathbf{E}_{\text{<REAL>}}, \mathbf{E}_{\text{<AIGEN>}})$ (Sec. 3.1) is realized empirically: does training on a strict subset of modalities $\mathcal{M}_{\text{train}} \subsetneq \mathcal{M}$ yield a pair that detects AI-generated content in held-out modalities? Tab. 3 reports AUROC for every combination of $\mathcal{M}_{\text{train}}$ in $\{\text{single modality}\} \cup \{\text{leave-one-out}\} \cup \{\mathcal{M}\}$, evaluated on every modality. We stress at the outset what the table does and does not establish: the pair is modality-agnostic *by construction*, in that it can be evaluated on any modality the backbone accepts at input, but the direction it learns is only *partially* shared. We note three observations.

First, transfer is real but strongly asymmetric, and it depends far more on the target modality than on the source. Re-measuring the whole single-modality block with confidence intervals (Tab. 12) resolves each cell into one of three regimes. Six of the twelve off-diagonal cells are above chance, the strongest being video $\rightarrow$ audio (0.845 [0.80, 0.89]) and text $\rightarrow$ image (0.706 [0.68, 0.73]). Two sit at chance (text $\rightarrow$ video, image $\rightarrow$ video). The remaining four—text $\rightarrow$ audio (0.323), video $\rightarrow$ text (0.331), audio $\rightarrow$ video (0.401), video $\rightarrow$ image (0.414)—are significantly *below* chance, with intervals that exclude 0.5 by a wide margin. Read as targets, no modality transfers to video at all, whereas audio and image are reached from two sources each. We therefore do not claim a single universal real-vs-AI direction: what the pair learns is partly shared and partly modality-specific, and in four source–target pairs it is actively misleading.

The sub-chance cells deserve a separate word, because they are not measurement noise. Their intervals are tight and far from 0.5, so the trained direction is genuinely anti-correlated with the label in those pairs—a real-vs-AI axis learned on one modality can order another modality backwards. This is a stronger and less comfortable statement than the one we made during review, where we attributed sub-chance entries to prompt-template noise; that attribution was based on a text checkpoint trained at a fraction of the documented data scale (App. C.3), and it does not survive re-measurement.

Second, leave-one-out training closes most of the gap on the held-out modality when text is in the training mixture (e.g., $\{\text{audio, image, video}\} \rightarrow \text{text}$ reaches 0.73, well above any single-modality transfer to text), but adding modalities without text does not (e.g., $\{\text{text, image, video}\} \rightarrow \text{audio}$ falls below the text-only transfer). This is consistent with text contributing the largest share of cross-modal signal to the shared embedding rows.

Third, joint training across all four modalities produces the most balanced performance: text, audio, and video are within 0.03 AUROC of their dedicated single-modality runs, and the only notable gap is on image (0.68 vs. 0.92). This gap is an artifact of the training schedule rather than of the shared representation. Our round-robin scheduler exhausts each modality’s loader, so at the dataset caps of Tab. 1 the all-four run takes roughly 20,000 text steps against about 250 image batches per epoch, giving image under 2% of the gradient signal. Rebalancing the budgets recovers most of the

Table 3: **Cross-Modality Generalization.** AUROC for every choice of $\mathcal{M}_{\text{train}}$ (single-modality, leave-one-out, all four), tested for ability to generalize to modalities unseen during training time. Values are single-seed runs at the evaluation scale of the submitted version and carry no intervals; the single-modality block is superseded by Tab. 12, which re-measures it at the revision’s scale with bootstrap CIs and classifies each cell against chance. Several entries move materially under that re-measurement (Sec. 4.3).

Train				Test			
Txt	Aud	Img	Vid	Txt	Aud	Img	Vid
✓				0.96	0.71	0.57	0.58
	✓			0.52	0.99	0.58	0.41
		✓		0.61	0.77	0.92	0.46
			✓	0.48	0.45	0.58	0.83
	✓	✓	✓	0.73	0.99	0.79	0.78
✓		✓	✓	1.00	0.48	0.81	0.80
✓	✓		✓	1.00	1.00	0.60	0.80
✓	✓	✓		0.56	0.99	0.41	0.47
✓	✓	✓	✓	0.99	0.99	0.68	0.80

Table 4: **Image Detection Robustness.** AUROC on the in-distribution GenImage test split under JPEG compression, bilinear resize-and-restore, and Gaussian blur, contrasted with the baseline. Our row is re-scored with percentile-bootstrap 95% CIs ($n = 500$); every value reported in the submitted version falls inside its interval.

Method	Clean	JPEG quality				Resize-restore			Gaussian blur (σ)		
		90	75	50	25	75%	50%	25%	0.5	1.0	2.0
UnivFD	0.56	0.54	0.54	0.54	0.55	0.58	0.57	0.58	0.55	0.59	0.55
AIDE	0.82	0.82	0.80	0.79	0.79	0.82	0.77	0.69	0.84	0.79	0.69
Gemma-4-E4B-AI-Neo	0.91	0.89	0.86	0.81	0.75	0.80	0.69	0.60	0.90	0.75	0.61
95% CI	[.89,.94]	[.86,.92]	[.83,.89]	[.77,.85]	[.71,.79]	[.76,.84]	[.65,.74]	[.55,.65]	[.87,.92]	[.70,.79]	[.57,.66]

Table 5: **Text Detection Robustness.** AUROC on held-out RAID source IDs, evaluated by attack category. Detection-token methods are trained only on the standard random training split. Our rows are re-scored with percentile-bootstrap 95% CIs; re-scoring moves the submitted values by 0.014 on average and leaves the ordering unchanged.

Method	Clean	Paraphrase	Misspelling	Whitespace	Casing	Homoglyph
DivEye	0.91	0.83	0.89	0.79	0.82	0.93
Binoculars	0.85	0.93	0.84	0.81	0.83	0.81
Llama-3.1-8B-AI-Neo	0.99	0.97	1.00	1.00	0.99	1.00
95% CI	[.99,1.00]	[.97,.97]	[.99,1.00]	[1.00,1.00]	[.99,1.00]	[1.00,1.00]
Gemma-4-E4B-AI-Neo	0.98	0.85	0.98	0.97	0.97	0.95
95% CI	[.98,.98]	[.84,.86]	[.98,.98]	[.96,.97]	[.97,.98]	[.94,.95]

difference: an unbalanced mini-replica reaches 0.77 on image, a balanced-budget joint run 0.78, and an image-only reference at matched scale 0.88 (App. C.3). Consistently with this, App. C.4 finds that pairs trained independently per modality occupy near-orthogonal directions, so what a joint run learns is a compromise between four largely distinct directions rather than one direction that all four modalities share.

4.4 Detection degrades gracefully under common perturbations

We evaluate our trained pair under modality-appropriate perturbations of varying strength on the in-distribution test sets, comparing against the strongest baseline per modality from Sec. 4.1.

Image perturbations. We apply (i) JPEG compression at quality factors $\{90, 75, 50, 25\}$, (ii) bilinear resizing to $\{75\%, 50\%, 25\%\}$ of the original resolution before re-upsampling, and (iii) Gaussian blur with $\sigma \in \{0.5, 1.0, 2.0\}$. Tab. 4 reports AUROC at each perturbation level. Aggressive bilinear resize is the worst case for both our pair and AIDE, while JPEG compression and mild Gaussian blur produce only modest degradations. Our pair starts **0.09** and **0.35** AUROC points above AIDE and UnivFD under clean conditions, and the gap narrows under the strongest perturbations.

Text perturbations. We adopt the attack categories defined by RAID: (i) paraphrasing, (ii) misspelling, (iii) whitespace insertions, (iv) casing perturbations, and (v) homoglyph substitutions, evaluating on the held-out RAID source IDs. Tab. 5 shows AUROC under each text perturbation. Our Llama-3.1-8B-AI-Neo pair is essentially flat across all five attack categories. The smaller Gemma-4-E4B-AI-Neo backbone is more sensitive to paraphrasing ($0.98 \rightarrow 0.85$) but holds up under the four character-level attacks. Neither text baseline matches our pair on any attack category, and the relative gap is preserved or widened under perturbation.

4.5 Comparison with MLLM-based detectors out of distribution

A separate body of recent work fine-tunes MLLMs end-to-end for AI-generated content detection, typically with human-annotated ra-

Table 6: **Comparison Against MLLM Detectors.** Out-of-distribution AUROC across modalities, using each method’s released checkpoint without further adaptation. N/A marks unsupported modalities. The **Mean** column reports coverage across modalities, not a head-to-head comparison: the MLLM baselines apply only to image. Baselines are published point estimates; CIs are available only for our rows.

tionales as supervision [Zhang et al., 2024, Li et al., 2025b, Xu et al., 2024, Huang et al., 2025b, Hondru et al., 2026]. Direct in-distribution comparison is not meaningful since each method is trained on its own dataset and we cannot retrain their checkpoints, so we instead compare on OOD datasets that none of the methods have seen during training, using each method’s released checkpoint.

Tab. 6 reports AUROC on the OOD benchmarks from Sec. 4.2—Beemo, In-the-Wild, CIFAKE, and DeeptraceReward—for SIDA [Huang et al., 2025b], FakeShield [Xu et al., 2024], and our trained pair. Both MLLM

baselines are image-only by construction, so the head-to-head comparison reduces to a single dataset, CIFAKE. There, our Gemma-4-E4B-AI-Neo pair reaches 0.61 [0.58, 0.63] AUROC at $n = 2,000$; FakeShield’s 0.61 falls inside that interval, so the supported statement is parity with the strongest MLLM-based detector on this transfer, alongside margins of +0.44 over SIDA, +0.14 over UnivFD, and +0.07 over AIDE. Parity is nonetheless obtained while training roughly seven orders of magnitude fewer parameters and using no human-annotated rationales. Sec. 4.2 further shows that CIFAKE conflates generator shift with a 32×32 content-domain shift; on the generator axis alone the same pair reaches 0.92 and 0.89. On the three modalities the MLLM baselines do not cover, the same pair attains 0.77 on text, 0.70 on audio, and 0.87 on video, for a four-modality mean of 0.74; Llama-3.1-8B-AI-Neo reaches 0.79 on text. That mean is a statement about coverage rather than a head-to-head comparison: the MLLM baselines cannot be applied to those three modalities at all.

4.6 Self-verbalizations are faithful

Verbalized words score consistently with the detector. We elicit, from the trained model, lists of words that are characteristic of <AIGEN> text and lists characteristic of <REAL> text. Denote these word sets W_{AI} and W_{human} . We then score each word w in isolation through the detector by computing $s_{\text{text}}(w)$ (Eq. 1). A faithful verbalization should satisfy $\mathbb{E}_{w \in W_{\text{AI}}}[s_{\text{text}}(w)] > \mathbb{E}_{w \in W_{\text{human}}}[s_{\text{text}}(w)]$. Practically, words the model identifies as AI-likely should themselves score AI-likely under the detector. As an untrained baseline, we elicit analogous word lists from the same backbone with the trained tokens removed, asking the model directly for AI-likely and human-likely words; we score these baseline words through the detector and compare. Tab. 15 reports the mean detector score for W_{AI} and W_{human} under both elicitation protocols across 480 samples per condition. The trained tokens produce a 0.91 log-probability gap in the expected direction; the natural-language baseline produces a gap of -0.08 in the wrong direction (one-sided Welch $p \approx 0$ for both). Equivalently, 54.2% of words in W_{AI} score AI under the detector, vs. 27.6% of baseline AI-likely words.

Plug-in evaluation. We further evaluate plug-in faithfulness in the sense of Hewitt et al. [2025b]: we replace the trained $\tau \in \{\text{<REAL>, <AIGEN>}\}$ in the prompt template Π_{text} with the natural-language verbalization $\nu(\tau)$ produced by the model itself, hand the resulting plain-text prompt to a frontier MLLM that has never seen our trained tokens, and ask the evaluator to decide, over a held-out pair $(x_{\text{real}}, x_{\text{AI}})$, which input is AI-generated. As baseline, the same evaluator sees the same pairs with $\nu(\tau)$ omitted, asked to classify real vs. AI directly. Tab. 7 reports pairwise accuracy: both Gemini 3.1 Pro ($0.47 \rightarrow 0.72$) and Gemini 3 Flash ($0.48 \rightarrow 0.75$) move from near-chance baselines to clearly above chance once the verbalization is plugged in, with Δ of +0.25 and +0.27 respectively. The model’s free-form description of <AIGEN>

Table 7: **Plug-in Evaluation.** Pairwise AI-vs-real classification accuracy on held-out text pairs, when the trained <AIGEN> embedding is replaced with its verbalization $\nu(\text{<AIGEN>})$ and handed to an off-the-shelf MLLM evaluator that has never seen our trained tokens.

Evaluator	Baseline	+ $\nu(\text{<AIGEN>})$	Δ
Gemini 3.1 Pro	0.47	0.72	+0.25
Gemini 3 Flash	0.48	0.75	+0.27

Text

<AIGEN>

1. Brevity: Short sentences and paragraphs, often with a focus on conveying a single idea or message. **2. Conciseness:** Avoidance of unnecessary words, phrases, or descriptions, with a focus on clarity and efficiency. **3. Directness:** Straightforward and unambiguous language, with minimal use of metaphors, similes, or other literary devices. **4. Objectivity:** A neutral or detached tone, often lacking emotional appeals or personal opinions. **5. Clarity:** Easy-to-understand language, with a focus on ...

<REAL>

1. Complex syntax: The text may have long, convoluted sentences with multiple clauses and phrases, making it hard to follow. **2. Technical jargon:** The text may use specialized vocabulary or technical terms that are not familiar to the average reader. **3. Abstract concepts:** The text may deal with abstract ideas or complex concepts that require a high level of cognitive processing to understand. **4. Lack of clarity:** The text may be written in a way that is ambiguous or unclear ...

Image

<AIGEN>

1. Geometric and Sharp: often features clean lines and precise angles. The shapes are rarely organic or blurry, instead leaning towards a structured, geometric aesthetic. **2. Minimalist and Sparse:** images tend to have a lot of negative space. The composition is often simple, with very few elements competing for attention. **3. Monochromatic or Limited Palette:** often uses a restricted color scheme, frequently relying on black, white, and shades of grey. ...

<REAL>

1. Abstract and Geometric: The images often feature intricate geometric patterns, abstract shapes, and complex lines, giving them a highly stylized and non-representational look. **2. High Contrast and Bold Colors:** There's a frequent use of bold, vibrant colors set against stark backgrounds or high-contrast combinations, creating a visually striking and energetic feel. **3. Digital and Glitchy:** Many images incorporate digital aesthetics, such as pixelation, glitch effects, and digital ...

Figure 2: **Global self-verbalizations of <REAL> and <AIGEN>.** Greedy completions of “<TOK> <modality> are characterized by...” for text (left) and image (right), decoded from a frozen MLLM whose only learnable parameters are the two embedding rows.

therefore carries enough detection signal for an unrelated, off-the-shelf MLLM to use, indicating that the verbalization is faithful to the embedding rather than fluent confabulation.

4.7 What the neologisms describe

We complement the quantitative faithfulness results with qualitative analysis of what the trained tokens describe. We elicit three kinds of self-verbalizations: *global* (concept-level, no input), *per-generator* (one neologism trained per generator), and *instance* (datapoint-conditioned).

Global verbalizations identify modality-appropriate regularities. Fig. 2 shows representative completions of “<AIGEN> <modality> are characterized by...” across the four modalities. The completions cite modality-appropriate regularities—e.g., conciseness and detached objectivity for AI-generated text; limited color palettes and composition for AI-generated images—all decoded from an otherwise-unchanged MLLM that has never been trained to articulate them.

Per-generator neologisms surface generator-specific tells. On EvoBench, we additionally train a separate <AIGEN_g> embedding for each generator *g*, holding the data and protocol identical to Sec. 4.2. Fig. 3 shows greedy verbalizations of each <AIGEN_g> elicited under the contrastive pairwise setup with <AIGEN_g> contrasted with samples from a different generator. Descriptions reflect known stylistic differences: *raw, unformatted text* for Gemini; *clear, unadorned phrasing* for Claude; a colloquial register for GPT-4o. The contrast requires the pairwise setup. Under the leave-one-generator-out protocol of Sec. 4.2 any two folds share about 90% of their training data and the resulting tokens are near-identical, which is consistent with the shared <AIGEN> embedding tracking discourse-level regularities that hold across generator families while a per-generator token supplies the fingerprint.

Instance verbalizations point at features in the specific input. A key affordance of training ($E_{\text{<REAL>}}$, $E_{\text{<AIGEN>}}$) inside a multimodal LM is that the same trained tokens can be probed conditional on an input. For a given x we prompt the model to explain why *this particular x* is or is not <AIGEN> (Sec. 3.2). Fig. 4 (in App. C.8) shows representative examples paired with the input, the detector prediction $\hat{y}(x)$, the score $s_m(x)$, and the elicited verbalization. *Their scope should not be overstated. Examining matched success and failure cases (App. C.9), we find that an instance verbalization mostly describes what the input contains rather than what makes it AI-generated: a*

Gemini vs. GPT-4o

<AIGEN_GEMINI>	
1. Raw or unformatted: The text is likely to be unedited, without proper punctuation, grammar, or spelling corrections. It may also be written in a plain or simple font without any formatting, such as headings, paragraphs, or bullet points.	
2. Unedited or uncensored: The text may contain profanity, slang, or other forms of language that would not be suitable for general audiences...	

Claude Sonnet vs. GPT-4o

<AIGEN_CLAUDE>	
written in a clear, concise, and straightforward manner.	
It is implied that the text in question is unadorned, possibly lacking in flair or eloquence, and may not be written in an overly complex or ornate style. The connotation is of a straightforward and factual communication, perhaps with a sense of simplicity and directness.	

GPT-4o vs. Gemini

<AIGEN_GPT4o>	
• Is likely to be informal or conversational	
• Is possibly written for a younger audience	
• May be written in a casual tone	
• Possibly includes colloquial expressions or slang	
• Could be used in a chat or messaging application	

Figure 3: **Per-generator self-verbalizations on EvoBench.** Greedy completions of <AIGEN_g> under three pairwise contrastive splits, with a separate token trained to discriminate between generators

correctly detected AI abstract and a correctly rejected human abstract both elicit “highly formal, academic register”, despite scores of +8.69 and −8.06. An individual verbalization is therefore not a per-decision explanation. What the cases do expose reliably is the shape of the decision boundary—text false positives are human-written encyclopedic prose, text false negatives are AI text written as casual dialogue, and image false positives are real photographs low in fine texture—and a usable confidence signal, since verbalized hedging tracks $|s_m(x)|$ and hence correctness.

We did not train the model to produce any of these descriptions; they are decoded from a frozen backbone whose only modification is two embedding rows. Taken individually the descriptions can read as unsurprising, and we do not claim otherwise. What makes them non-trivial is that there is no cheaper way to obtain them: the decision direction $\delta = E_{\text{<AIGEN>}} - E_{\text{<REAL>}}$ has no lexical neighborhood at all—its highest cosine to any vocabulary token is 0.07–0.15 across text, image, and video checkpoints, at or below the random-vocabulary baseline, and the tokens attaining it are semantically empty fragments (App. C.4). No word in the vocabulary captures the learned axis, so decoding it through the frozen backbone into content that transfers detection signal to a third-party model (Tab. 7) is the substantive claim, not the vividness of any one description. The most surprising finding is instead an asymmetry: the words the model volunteers for <AIGEN> are structural and register-based (“introduction”, “abstract”, “conclusion”), not the stylistic tells human readers report (“however”, “furthermore”), which in fact score *below* the human-written baseline (Tab. 15). We therefore take Figs. 2–4 as evidence that the trained tokens have acquired **content-grounded, decodable** semantics rather than an opaque discriminative direction. Additional examples, including failure modes and per-generator comparisons on EvoBench [Yu et al., 2025], are in App. C.8.

5 Discussion

To our knowledge, this is the first AI-generated content detector to span text, image, audio, and video through a single shared decision representation. Prior cross-modality work either fuses synchronous audio-visual streams [Khalid et al., 2021, Ilyas et al., 2023, Bohacek and Farid, 2024b] or stacks per-modality heads on a shared encoder [Huang et al., 2025a], neither of which can be applied to a held-out modality at all. What our results support is narrower than a universal real-vs-AI direction: the pair is modality-agnostic by construction, and empirically the direction is partially shared—above chance for several source–target pairs, and chance for others (Sec. 4.3). The embedding geometry says why. Pairs trained independently per modality occupy directions whose mutual cosines sit at or below the random-vocabulary baseline (App. C.4), so what a single modality learns is largely its own axis, and a joint run trades those axes off against one another rather than finding one they share. We read the practical implication as favoring joint training over transfer: the cost of adding a modality is two rows, not a new head. The same edit furnishes an interpretability handle that prior explainable detectors purchase only with human-annotated rationales [Zhang et al., 2024, Liu et al., 2024a, Huang et al., 2025b, Hondru et al., 2026]: a practitioner can query the trained tokens in free-form natural language—globally, per generator, or conditional on a specific input—and read out what the detector encodes, validated via plug-in evaluation. With a trainable footprint on the order of $2d$ parameters and verbalization intrinsic to the backbone, the recipe should extend cheaply to new modalities (e.g., 3D, motion capture), per-generator diagnostics, and, more speculatively,

other binary decisions whose boundary admits a natural-language description. The same footprint also changes how one responds to a new generator: rather than retraining a detector, a practitioner can re-fit two rows on a small labeled sample—1,000 pairs suffice to move CIFAKE from 0.61 to 0.97 and OpenFake from 0.64 to 0.97 (App. C.5)—or maintain a library of per-generator <AIGEN_{*g*}> tokens as diagnostics.

Limitations. We begin with the failure modes of the method itself. *Transfer is not universal, and can invert.* A pair trained on one modality does not reach every other one: no modality transfers to video above chance, and in four source–target pairs the learned direction is significantly *anti*-correlated with the label rather than merely uninformative (Tab. 12). The embedding geometry of App. C.4 suggests why—per-modality pairs are close to orthogonal, so there is no single direction to inherit, and an arbitrary projection of one onto another is as likely to be negative as positive. *Joint training is schedule-sensitive.* The image drop under all-four training is a gradient-share artifact of round-robin scheduling (Sec. 4.3), which means the method inherits a scheduling hyperparameter that a per-modality head would not have. *Everything is bounded by the backbone.* The method requires a backbone that already encodes every target modality, so it cannot reach a modality the backbone does not accept; verbalization quality is likewise bounded by the backbone’s own articulation ability, which is visibly weaker for audio than for text or image (App. C.9). *Cheap to train is not cheap to run.* The trainable footprint is $2d$ scalars, but inference is a full MLLM forward pass per prompt—two per example, given the paired template—so the method is far more expensive at test time than the small dedicated detectors it is compared against (App. A.6). *Verbalizations are not per-decision explanations.* As Sec. 4.7 reports, an instance verbalization frequently describes the input rather than the evidence for the decision. Though the proposed detection method retains strong performance across adversarial attacks, detection success varies with the generator model. As a result, there may be generator models trained in the future that degrade detection performance. More broadly, any AI-generated content detector may produce both false positives and false negatives, particularly under distribution shift. We therefore do not recommend deployment in high-stakes settings without human oversight and additional testing in the target domain. Misusing AI-detection tools can degrade trust in reliable sources including media organizations, students, and institutions, while incorrectly granting authenticity to deceptive or synthetic content that evades detection. With regard to natural-language descriptions, we make a best-effort attempt to evaluate faithfulness. However, true counterfactual tests are not possible in our setting as counterfactuals would themselves be either AI- or human-written, and thus confound evaluation. Our hope is that verbalizations provide insight into the mechanisms behind AI-detection, but they should not be interpreted as holistic causal explanations of the model.

Future work. The most immediate direction is breadth of backbone: we validate the recipe on four (App. C.1), all decoder-only and three from one vendor family, and a systematic study across architectures—in particular ones whose modality encoders are trained differently—would test how much of the shared-direction behavior is a property of the recipe versus of how a given backbone aligns its modalities.

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A Implementation Details

A.1 Hyperparameter Selection

For each backbone, we selected hyperparameters using a small held-out validation split of 1k RAID examples. For Llama-based neologisms, we train with batch size 32, and learning rate 10^{-4} . For Gemma-4-E4B-it, we used the same split and evaluation protocol, but selected a larger learning rate, 10^{-3} . In each case, we warmup for 20 steps and then cosine-decay to a minimum learning rate 0.01x smaller. For both models, we use the AdamW optimizer.

Training objective. All results reported in this paper were produced with the cross-entropy objective of Eq. 3; the main-text description of the method is what was run. An earlier version of this work optimized the APO-up objective of Hewitt et al. [2025b] instead, and a previous revision of this appendix described that objective in error. We restate it here because we report an ablation against it. Hewitt et al. [2025b] select APO/DPO over a likelihood objective on the grounds of improved *steering* behavior; since we do not evaluate steering, we tested both in our setting and found them to perform comparably, with cross-entropy being more straightforward to describe. The APO-up objective is

$$\mathcal{L}(y_c, y_r) = -\log \sigma \left(\beta \log \frac{p_\theta(y_c)}{p_\theta(y_r)} + \beta \log \frac{p_{\theta_0}(y_c)}{p_{\theta_0}(y_r)} \right) - \alpha_{\text{APO}} \log \sigma \left(\beta \log \frac{p_\theta(y_c)}{p_{\theta_0}(y_c)} \right), \quad (6)$$

where p_θ is the model with the trainable neologism embedding, p_{θ_0} is the frozen reference, y_c is an AI-generated response, and y_r is a non-AI-generated response (or vice versa for the <REAL> token). In all cases we condition on the fixed prompts specified in the body, differing from the original implementation that contained per-response instructions. We set $\beta = 0.1$ and $\alpha_{\text{APO}} = 1.0$.

Tab. 8 reports both objectives on Gemma-4-E4B-it across all four modalities. The two are within overlapping confidence intervals on every modality, which is why we report the simpler one throughout.

Table 8: **Training Objective Ablation.** AUROC on Gemma-4-E4B-it under the cross-entropy objective of Eq. 3, which produced every result in this paper, and under the APO-up objective of Hewitt et al. [2025b]. The two agree within overlapping 95% CIs on all four modalities.

Modality	Evaluation set	APO-up	Cross-entropy
Text	RAID	0.96 [0.96, 0.97]	0.96 [0.95, 0.96]
Audio	AUDETET + In-the-Wild	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]
Image	GenImage	0.84 [0.77, 0.91]	0.87 [0.81, 0.93]
Video	DeepraceReward	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]

A.2 Gemma 4 Adaptations

The auxiliary per-layer embedding table. Most decoder-only LMs maintain a single learnable embedding matrix $\mathbf{E} \in \mathbb{R}^{d \times |\mathcal{V}|}$ from which token vectors are read at the input and (via tied weights) the output. Gemma 4 E-series carries a *second* token-indexed table, the per-layer embeddings $\mathbf{E}^{\text{PLE}} \in \mathbb{R}^{h \times |\mathcal{V}|}$, whose rows are injected into each transformer layer as a layer-specific auxiliary signal alongside the residual stream. Vocabulary expansion (Sec. 2) therefore requires resizing *both* tables: we extend $\mathbf{E}$ with two trainable rows ($\mathbf{E}_{\langle \text{REAL} \rangle}$, $\mathbf{E}_{\langle \text{AIGEN} \rangle}$) as described in the main text, and we extend $\mathbf{E}^{\text{PLE}}$ with two rows initialized to the per-vocab mean $\frac{1}{|\mathcal{V}|} \sum_v \mathbf{E}_v^{\text{PLE}}$.

Why $\mathbf{E}^{\text{PLE}}$ is held frozen. The new PLE rows are not trained. We found that allowing gradients to flow into both tables lets the optimizer distribute the discriminative signal across them, leaving the rows of $\mathbf{E}$ close to their warm-start initialization. Because the LM head is tied to $\mathbf{E}$, this collapses subsequent self-verbalization (Sec. 3.2) to the lexical neighborhood of the warm-start word: probes elicit near-synonyms of the initialization token rather than descriptions of the learned concept. Freezing $\mathbf{E}^{\text{PLE}}$ at the mean-fill values forces all discriminative signal into the rows of $\mathbf{E}$ and preserves the verbalization channel. The trainable parameter count (Sec. 3.1) is unchanged at $2d$.

Gradient masking. On backbones with tied input/output embeddings, an unrestricted gradient on E would also update every other vocabulary row through the LM-head loss. We register a hook on $E.\text{grad}$ that zeros all rows except those of $\langle \text{REAL} \rangle$ and $\langle \text{AIGEN} \rangle$; combined with E^{PLE} being frozen, this guarantees that only the two new rows of E receive non-zero updates over the course of training.

Per-modality prompt templates. The image and text templates of Sec. 3.1 place τ at two positions; for audio and video we use the simpler single-position forms $\Pi_{\text{audio}}(\tau, \mathbf{x}) = \langle \text{audio}(\mathbf{x}) \rangle$ “This is a τ ” and $\Pi_{\text{video}}(\tau, \mathbf{x}) = \langle \text{frames}(\mathbf{x}) \rangle$ “These frames are from a τ video”. Video frames are sampled uniformly (default eight frames per clip) and passed through the multimodal processor as a sequence of image placeholders; audio waveforms are resampled to 16 kHz and clipped to 25 s before being handed to the processor’s audio path.

Warm-start tokens. Sec. 3.1 initializes both new rows from a neutral existing token. In practice we request a modality-appropriate noun: photograph (image), audio (audio), writing (text), video (video). [One caveat applies to the image runs. photograph is not a single vocabulary item in either Gemma tokenizer, and the resolution step falls back to text when a requested word does not map to a single id, so every image run reported here—in the submitted version and in this revision—was in fact warm-started from text.](#) We report this because it is visible in the released code and because it explains why the raw nearest neighbors of the image rows are translations of *text* (App. C.4). It does not affect any reported number: retraining image seed 42 from a genuine single-token image noun (photo) and from an explicit text control leaves ID AUROC within 0.01 of the as-run 0.90, consistent with the warm-start ablation in App. C.3. The fallback is no longer silent, and each run now records its resolved warm start. These are not semantically inert — they carry modality-relevant content — but they place the optimizer at an initialization already aligned with the surrounding context, so post-training divergence between $E_{\langle \text{REAL} \rangle}$ and $E_{\langle \text{AIGEN} \rangle}$ reflects the real-vs-AI direction rather than coarser modality information that would otherwise have to be re-learned. For the video modality the warm-start token coincides with the continuation token (Sec. 3.1); we did not observe this to interfere with training, but a non-overlapping warm-start (e.g. footage) is a sensible alternative.

A.3 Continuation-Token Selection

The score $s_m(\mathbf{x})$ in Eq. 1 is evaluated against a fixed continuation token $w_m \in \mathcal{V}$ that names the modality in plain English. Its role is to pin a concrete completion under both prompts so the discriminative signal isolates to the difference between $E_{\langle \text{REAL} \rangle}$ and $E_{\langle \text{AIGEN} \rangle}$. The choice of w_m is therefore not unique: any token that the backbone treats as a plausible completion of the prompt template will serve. The only hard requirement is that w_m be a single id in $\mathcal{V}$, since a multi-token continuation would either require summing log-probabilities across positions or arbitrarily picking one position to score.

In practice we resolve w_m at runtime by walking an ordered list of candidates per modality and selecting the first that maps to a single id distinct from the unknown token. The lists are

w_{image} : Image, image, photo, photograph
 w_{audio} : Audio, audio, recording, voice, sound, clip
 w_{video} : video, Video, recording, footage, clip
 w_{text} : text, Text, writing, passage, content

For the Gemma 4 tokenizer, the first candidate in each list resolves to a single id and is the one used throughout our experiments; the remaining candidates are retained as backbone-portable fallbacks for tokenizers where the leading capitalized form does not exist as a single piece.

Sensitivity to the choice of w_m . Because the choice is not unique, we measure how much it matters. We re-score the trained image and text checkpoints under every candidate in the lists above, changing nothing else; Tab. 9 reports the result. In distribution the score is essentially independent of the continuation token: the image spread across five candidates is 0.021 AUROC with every interval overlapping, and the text spread is 0.0002. Under distribution shift the choice matters more (0.035 on image, 0.095 on text), which is what one would expect if w_m selects how the frozen backbone

phrases the question and that phrasing carries more weight the further the input is from anything seen in training. The defaults are not cherry-picked: picture beats Image on CIFAKE, and writing and passage beat text on Beemo. The same sensitivity, read the other way, explains the swing observed on the text $\rightarrow$ image transfer cell in App. C.3: token choice dominates precisely where there is no signal for it to interact with.

Table 9: **Continuation-Token Robustness.** AUROC of a fixed trained checkpoint re-scored under each candidate continuation token w_m , $n = 500$ per split. In-distribution detection is nearly independent of the choice; out-of-distribution it is not. Defaults used throughout the paper are marked $\dagger$.

w_{image}	ID	OOD	w_{text}	ID	OOD
Image $\dagger$	0.91	0.61	text $\dagger$	1.000	0.60
image	0.90	0.59	Text	0.9998	0.55
photo	0.89	0.59	writing	1.000	0.65
picture	0.89	0.62	passage	1.000	0.64
art	0.89	0.61	content	1.000	0.58
Spread	0.021	0.035	Spread	0.0002	0.095

A.4 Baselines

We validate our proposed approach by comparing it against several state-of-the-art baselines.

Text. We compare against DivEye¹ [Basani and Chen, 2025], Binoculars² [Hans et al., 2024], and TMR-Detector Oxidane [2025]. DivEye is a supervised text detector that extracts a 10-dimensional feature vector from each sample using the default language model, GPT-2. The features are surprisal-based statistics: the mean, standard deviation, variance, skewness, and kurtosis of per-token surprisal; the mean and standard deviation of first-order surprisal differences; and variance, entropy, and auto-correlation of second-order log-likelihood differences. These features are then fed into an XGBoost classifier. The XGBoost hyperparameters are unchanged. Binoculars is a zero-shot text detector that computes a cross-perplexity ratio between a base model (Falcon-7B) and its instruction-tuned counterpart (Falcon-7B-Instruct). Lower scores indicate AI-generated text. Since no training is required, we kept the underlying model and scoring logic the same as the original. TMR-Detector is the highest-scoring detector by AUROC on the private RAID test set. It is a RoBERTa-based classifier trained on RAID using a hard-example mining strategy. Lastly, we evaluate linear classifiers trained directly on the final last-layer activation of Llama-3.1-8B and Gemma-4-E4B-it, reported in Tab. 1 as Llama-3.1-8B-Classifier and Gemma-4-E4B-Classifier.

Audio. We compare against AASIST [Jung et al., 2022] and RawNet2 [Tak et al., 2021], using the same repositories³. Both of these models are end-to-end audio anti-spoofing models. AASIST uses integrated spectro-temporal graph attention networks, and RawNet2 operates directly on raw waveforms.

Image. We compare against UnivFD⁴ [Ojha et al., 2023] and AIDE⁵ [Yan et al., 2024]. UnivFD uses a frozen CLIP ‘ViT-L/14’ as a feature extractor and trains only a single linear classification layer on top. The original paper trained on ProGAN images. For our purposes, we train the linear head on the GenImage dataset. The CLIP backbone and linear head architecture are unchanged. AIDE fuses features from a ‘ResNet-50’ and a ‘ConvNeXt’ backbone to capture both visual artifacts and noise-level patterns. The model architecture and training code are unchanged from the original repository.

Video. We compare against SBI⁶ [Shiohara and Yamasaki, 2022] and CDFA [Lin et al., 2024]. SBI trains a frame-level ‘EfficientNet-B4’ with a self-blending data augmentation strategy where real

¹<https://github.com/IBM/diveye/tree/main>

²<https://github.com/ahans30/Binoculars>

³<https://github.com/clovaai/aasist>

⁴<https://github.com/WisconsinAIVision/UniversalFakeDetect>

⁵<https://github.com/shilinyan99/AIDE>

⁶<https://github.com/mapooon/SelfBlendedImages>

frames are converted into pseudo-fakes by alpha-blending a color-jittered and affine-warped copy of the frame back into itself. The original paper uses facial landmark detection to constrain blending to face regions. Since our video detection task does not assume face content, we implemented a landmark-free analog, a randomly-sized rectangular region with a soft elliptical alpha falloff is blended in, followed by optional JPEG re-compression. At evaluation time, eight evenly spaced frames are extracted per video and the per-video score is the mean of frame-level scores. CDFA is a frame-level deepfake detector using an Xception backbone, trained with frequency-domain augmentation to encourage the network to learn forgery artifacts orthogonal to the natural image manifold. Specifically, random DCT-band masking (randomly suppressing either low or high frequency components) combined with JPEG re-encoding is applied during training. Per-video scores are computed as the mean of frame-level scores over 8 evenly-spaced frames.

Multimodal performance. We compare against SIDA⁷ [Huang et al., 2025b] and FakeShield⁸ [Xu et al., 2024]. SIDA is a large multimodal model for social media image deepfake detection, localization, and explanation. It builds on LLaVa + LISA + SAM and trains in two stages: a classification stage using the ‘[CLS]’ token and a segmentation stage that predicts tampered region masks while producing natural language explanations. We use the pretrained ‘SIDA-13B’ checkpoint for inference. SIDA was trained on SID_Set, which consists of real images, tampered images, and fully synthetic images. FakeShield is a two-module multimodal framework for explainable image forgery detection and localization. The Domain Tag-guided Explainable Forgery Detection Module classifies the forgery type using domain tags, and the Multi-modal Forgery Localization Module generates pixel-level tampered region masks alongside natural language explanations. We use the pretrained ‘FakeShield-v1-22B’ checkpoint for inference.

A.5 Datasets

We evaluate our proposed approach across four modalities (text, audio, image, and video) using a three-set design for each: an in-distribution (ID) dataset with training and evaluation splits and an out-of-distribution (OOD) dataset used exclusively for held-out evaluation.

Text. For the ID dataset, we use RAID⁹ [Dugan et al., 2024], which is a large-scale binary AI-text detection benchmark containing texts produced by a diverse set of LLMs as well as human authors. We train on 20,000 samples (10,000 real/AI pairs) and evaluate on 6,000 held-out samples from RAID. For the OOD dataset, we utilize Beemo¹⁰ [Artemova et al., 2025], which provides paired prompt/response data with a human gold answer and an LLM response for the same prompt. Each row yields one human and one AI sample. We test on 4,326 samples.

Audio. For the ID dataset, we use In-The-Wild (ITW)¹¹ [Müller et al., 2022] and AUDETER¹² [Wang et al., 2025b]. ITW contains 37.9 hours of speech from 58 celebrity speakers (20.7 hours of real and 17.2 hours of AI-spoofed audio). We use only the bonafide (real) half for training. AUDETER is a large-scale audio deepfake corpus. We use the ‘audiobook-tts-cosyvoice’ subset (MLS audiobook speech synthesized with CosyVoice) for training fakes. The ITW real clips and the AUDETER fake clips is chosen to expose the model to modern synthesis methods while keeping ITW’s deepfakes as the unseen held-out set. Training uses 2,400 total samples (1,200 real/AI pairs), and in-distribution evaluation 2,000 samples (1,000 real/AI pairs). For the OOD dataset, we use ITW where the real rows are the complement of those used for training (excluded via a seeded stratified replay) and fake rows are ITW’s native deepfakes. This produces a clean both-class OOD split. We use 2,000 samples (1,000 real/AI pairs) for OOD testing.

Image. We use GenImage [Zhu et al., 2023b] for the ID dataset. We use a pre-processed mirror¹³ (224x224 JPEG) with a 70/15/15 train/val/test split. Training uses 1,000 samples (500 real/AI pairs)

⁷<https://huggingface.co/saberz/SIDA-13B>

⁸<https://huggingface.co/zhipeixu/fakeshield-v1-22b>

⁹<https://huggingface.co/datasets/liamdugan/raid>

¹⁰<https://huggingface.co/datasets/toloka/beemo>

¹¹<https://huggingface.co/datasets/mueller91/In-The-Wild>

¹²<https://huggingface.co/datasets/wqz995/AUDETER>

¹³<https://huggingface.co/datasets/Qwerty0193/genimage-processed>

and 2,000 samples (1,000 real/AI pairs) for evaluation. CIFAKE¹⁴ [Bird and Lotfi, 2024] is used as the OOD dataset, which contains 60,000 real CIFAR-10 images and 60,000 images generated by Stable Diffusion v1.4. It provides a strong distribution shift relative to GenImage’s multi-generator ImageNet-domain training images, testing resolution robustness and generator generalization simultaneously. 2,000 samples (1,000 real/AI pairs) are used for OOD testing.

Video. We use Deepfake-Eval-2024¹⁵ (DFE-2024) [Chandra et al., 2025] as the ID dataset. DFE-2024 contains in-the-wild deepfakes collected from social media in 2024, real-world adversarial samples drawn from 88 websites across 52 languages. It provides a harder, more realistic training distribution than lab-controlled benchmarks. The dataset comprises 1,221 training and 815 test videos. We use 1,000 samples (500 real/AI pairs) for training and [the complete 815-video test split for evaluation \(772 after class balancing, of which 768 decode\)](#). We use DeeptraceReward¹⁶ [Fu et al., 2025] as the OOD dataset. DeeptraceReward contains 6,636 balanced clips (3,318 real/AI pairs) spanning seven modern text-to-video generators. It acts as a controlled, named-generator counterpart to DFE-2024’s in-the-wild training distribution, testing whether features learned on real-world adversarial deepfakes generalize to specific known generators. [The complete held-out test split \(\$n = 664\$ \)](#) is used for OOD testing.

Sampling protocol and uncertainty. The remainder of this subsection documents exactly how every evaluation subset above is constructed, and how uncertainty is estimated.

Source splits. Where a dataset ships its own test split, we sample only from it and never from any other split: the GenImage mirror’s test split, CIFAKE’s test split, and DFE-2024’s 815-video test split. DeeptraceReward and Beemo are used as evaluation-only sources in full. For audio, the in-distribution row pairs In-the-Wild bonafide clips with AUDETER fakes, and the OOD row uses In-the-Wild’s native deepfakes against the *complement* of the real clips consumed in training, obtained by replaying the training-time sampling with the same seed so that no training example can reappear at evaluation.

Text: train/test document overlap. Text is the one modality where this exclusion is not applied, and the consequence is large enough that we report it explicitly. RAID does not ship a test split, so train, validation and in-distribution test are all drawn from its train table. Within a single request the three are a disjoint stratified partition; but training and in-distribution evaluation are issued as separate requests, and the second does not exclude the rows consumed by the first. That would be harmless if the table were large, and it is not: after the adversarial filter (attack = none) RAID’s train table contains exactly 1,766 unique human documents, each replicated across its 11 attack variants. The class-balanced pool is therefore the same 1,766 documents in both requests, and the in-distribution test set is essentially the training set.

We measured the size of the effect on a Llama-3.1-8B pair trained here under the protocol above. Scored on the RAID split sampled as described, it reaches 1.000 AUROC; scored on 6,000 balanced samples drawn from RAID’s disjoint extra table, which shares no documents with train, the same checkpoint reaches 0.560 [0.546, 0.575]. Every in-distribution text number in this paper should be read with that gap in mind. The extra table is a separate document pool rather than a random split of the same population, so 0.560 mixes unseen documents with some domain shift and bounds the leakage-free figure from below rather than isolating it. Sec. 4.3’s transfer matrix uses the extra pool for its text column for this reason; the released `prepare RAID heldout.py` builds it.

Subsampling rule. All subsets come from one deterministic procedure, with no manual selection, filtering, or inspection of any example before inclusion. Binary labels are resolved for every row of the source split; a class-balanced 50/50 down-sample to the requested size n is then taken by shuffling each class with a seeded Fisher–Yates pass and keeping the first $n/2$, after which the union is shuffled once more with the same generator. The base seed is 42 for every result in this paper. Where two evaluation rows draw from one candidate pool, the seed is offset by a hash of a per-row context key, so the rows are disjoint in expectation and still bit-exact reproducible.

¹⁴<https://huggingface.co/datasets/dragonintelligence/CIFAKE-image-dataset>

¹⁵<https://huggingface.co/datasets/nuriachandra/Deepfake-Eval-2024>

¹⁶<https://huggingface.co/datasets/DeeptraceReward/RewardData>

Sizes. The submitted version used $n = 100$ for image and video, ID and OOD. This revision reports image ID and OOD at $n = 2,000$, video ID on the complete DFE-2024 test split (772 videos after balancing; 768 scored, four clips being undecodable and logged as such), and video OOD on the complete held-out DeepraceReward split ($n = 664$). Text is evaluated at $n = 6,000$ (ID) and $n = 4,326$ (OOD), audio at $n = 2,000$.

Uncertainty. Every AUROC we report for our own models carries a percentile-bootstrap 95% confidence interval over test examples (10,000 resamples with replacement, fixed generator, one-class resamples skipped). Training-seed variability is reported separately as the standard deviation over three full training runs (seeds 42, 1, 2): on image, ID 0.892 ± 0.008 and OOD 0.582 ± 0.047 , i.e. in-distribution performance is seed-stable while OOD under the extreme CIFAKE shift is not; on text, ID and OOD both move by at most 0.01 for either backbone. Baseline entries throughout the paper are published point estimates obtained from released checkpoints, for which we do not hold per-sample scores and therefore cannot bootstrap.

Released artifacts. For every evaluation row we release a manifest listing scoring order, raw-row index, label, and per-sample detector score, together with the seeds and the preparation scripts. These reconstruct every evaluation set bit-exactly and let subsequent work compare against our per-sample predictions rather than only against a summary metric.

A.6 Engineering Efficiency

Because the paper claims parameter efficiency, we report what the recipe actually costs. All measurements are on a single H100-80GB (distinct from the hardware used for the experiments themselves; see App. A.7) with Gemma-4-E4B-it (7.94B parameters, $d = 2,560$, hence $2d = 5,120$ trained values), at batch size 1 and sequence length 512.

The backbone alone occupies 16.10 GiB in a forward-only pass. Peak memory during training reaches 23.39 GiB, and a training step takes 0.152 s against 0.081 s for a forward pass. At the dataset scale of Tab. 1, five epochs take about 1.5 hours for audio and 25 minutes for image. The trained artifact is two embedding rows, 12 KB on disk.

One honest qualification: what is *learned* is $2d$ values, but our implementation realizes the two rows by marking the whole $262,144 \times 2,560$ embedding table trainable and zeroing all other rows in a backward hook. The optimizer therefore receives 671M parameters and allocates gradient and moment buffers for the full table, which is where the gap between 16.10 and 23.39 GiB comes from. An implementation that splices a $2 \times d$ parameter into the embedding output would attain the memory cost the $2d$ figure implies; the numbers above measure the path we ran, not that idealization. This does not affect the parameter claim—the checkpoints contain exactly two rows—but it does mean the training-memory advantage over a small dedicated detector is smaller than the parameter count suggests. Inference cost is likewise unchanged from the backbone’s: two forward passes per example, one per member of the token pair.

A.7 Compute Resources

The experiments reported in this paper were conducted on two rented NVIDIA GH200 GPUs. Because our method updates only two embedding rows in an otherwise frozen backbone, training was computationally lightweight relative to standard MLLM fine-tuning. Individual training runs took roughly 2-10 hours depending on modality, base model, and training set size. Across all experiments reported in the paper, the total wall-clock compute time was approximately four days. Evaluation, including cross-modal transfer and perturbation experiments, was performed on the same hardware.

B Self-Verbalization Probes

This appendix documents the prompts and decoding protocol used to elicit self-verbalizations of $E_{\langle \text{REAL} \rangle}$ and $E_{\langle \text{AIGEN} \rangle}$ in Sec. 3.2 and Sec. 4.7.

B.1 Global Verbalization

Image.

User: *<AIGEN> is a concept used to classify images. Before you answer, describe what <AIGEN> images look like. I am trying to understand what <AIGEN> means to you.*

Assistant (forced start): *Ok, <AIGEN> images are visually characterized by ...*

User: *List 5 visual properties or descriptive phrases for the concept <AIGEN>. I am trying to understand what <AIGEN> means to you.*

Assistant (forced start): *Ok, here are 5 visual properties of <AIGEN> images: ...*

User: *Describe an image that perfectly embodies the concept <AIGEN>. What does it look like?*

Assistant (forced start): *An image that perfectly embodies <AIGEN> would ...*

Video.

User: *<AIGEN> is a concept used to classify videos. Before you answer, describe what <AIGEN> videos look like. I am trying to understand what <AIGEN> means to you.*

Assistant (forced start): *Ok, <AIGEN> videos are visually characterized by ...*

User: *List 5 visual or temporal properties or descriptive phrases for the concept <AIGEN>. I am trying to understand what <AIGEN> means to you.*

Assistant (forced start): *Ok, here are 5 visual/temporal properties of <AIGEN> videos: ...*

User: *Describe a video that perfectly embodies the concept <AIGEN>. What does it look like?*

Assistant (forced start): *A video that perfectly embodies <AIGEN> would ...*

Audio.

User: *<AIGEN> is a concept used to classify audio. Before you answer, describe what <AIGEN> audio sounds like. I am trying to understand what <AIGEN> means to you.*

Assistant (forced start): *Ok, <AIGEN> audio recordings are acoustically characterized by ...*

User: *List 5 acoustic properties or descriptive phrases for the concept <AIGEN>. I am trying to understand what <AIGEN> means to you.*

Assistant (forced start): *Ok, here are 5 acoustic properties of <AIGEN> audio: ...*

User: *Describe an audio recording that perfectly embodies the concept <AIGEN>. What does it sound like?*

Assistant (forced start): *An audio recording that perfectly embodies <AIGEN> would ...*

Text.

User: *<AIGEN> is a concept used to classify written text. Before you answer, describe what <AIGEN> text reads like. I am trying to understand what <AIGEN> means to you.*

Assistant (forced start): *Ok, <AIGEN> text is linguistically characterized by ...*

User: *List 5 linguistic or stylistic properties or descriptive phrases for the concept <AIGEN>. I am trying to understand what <AIGEN> means to you.*

Assistant (forced start): *Ok, here are 5 linguistic properties of <AIGEN> text: ...*

User: *Describe a piece of text that perfectly embodies the concept <AIGEN>. What does it read like?*

Assistant (forced start): *A piece of text that perfectly embodies <AIGEN> would ...*

B.2 Likely-Words Verbalization

User: Write words that are more common in <AIGEN> text as opposed to <HUMAN> text. One per line, no explanation.

Assistant prefill: Words more common in <AIGEN> text as opposed to <HUMAN> text:\n1.

For the human direction, we swap <AIGEN> and <HUMAN>. For the natural-language baseline, we replace them with “AI-generated” and “human-written”.

B.3 Instance Verbalization

Image.

User: <image> *The concept <AIGEN> is being used to classify this specific image. Describe precisely what you observe in this image — textures, edges, lighting, noise patterns, sharpness, skin detail, background coherence, or any subtle visual artifacts — that are most relevant to the concept <AIGEN>. Be concrete and specific to what you actually see here, not generalities.*

Assistant (forced start): *In this specific image, the visual characteristics most relevant to <AIGEN> are: ...*

User: <image> *You are evaluating this image under the concept <AIGEN>. List 3 precise visual cues you observe in this image — such as sharpness, color distribution, facial texture, reflection consistency, background blending, edge aliasing, or unusual smoothness — that align with or contradict <AIGEN>.*

Assistant (forced start): *Here are 3 specific visual cues I observe in this image that relate to <AIGEN>: ...*

User: <image> *Imagine explaining to a skeptical human why THIS specific image should be labeled <AIGEN> versus the alternative. What do you see in this image that supports or undermines the concept <AIGEN>? Be honest about any evidence on both sides.*

Assistant (forced start): *Looking at this specific image, <AIGEN> applies (or does not apply) because ...*

Video.

User: <frames> *The concept <AIGEN> is being used to classify this specific video. Describe precisely what you observe across these frames — textures, edges, lighting, motion consistency, facial detail, background stability, temporal coherence, or any subtle visual or temporal artifacts — that are most relevant to the concept <AIGEN>. Be concrete and specific to what you actually see here, not generalities.*

Assistant (forced start): *In these specific frames, the visual and temporal characteristics most relevant to <AIGEN> are: ...*

User: <frames> *You are evaluating this video under the concept <AIGEN>. List 3 precise visual or temporal cues you observe across these frames — such as sharpness, facial texture, motion smoothness, edge aliasing, frame-to-frame consistency, background coherence, or unusual artifacts — that align with or contradict <AIGEN>.*

Assistant (forced start): *Here are 3 specific cues I observe in these frames that relate to <AIGEN>: ...*

User: <frames> *Imagine explaining to a skeptical human why THIS specific video should be labeled <AIGEN> versus the alternative. What do you see across these frames that supports or undermines the concept <AIGEN>? Be honest about any evidence on both sides.*

Assistant (forced start): *Looking at these specific frames, <AIGEN> applies (or does not apply) because ...*

Audio.

User: <audio> *The concept <AIGEN> is being used to classify this specific audio clip. Describe precisely what you hear in this clip — prosody, intonation, breath sounds, formant quality, naturalness, emotional consistency, background noise, or any subtle acoustic or*

spectral artifacts — that are most relevant to the concept <AIGEN>. Be concrete and specific to what you actually hear here, not generalities.

Assistant (forced start): *In this specific audio clip, the acoustic characteristics most relevant to <AIGEN> are: ...*

User: *<audio> You are evaluating this audio clip under the concept <AIGEN>. List 3 precise acoustic cues you hear in this clip — such as prosody, formant clarity, breath naturalness, spectral artifacts, background ambience, or unusual smoothness — that align with or contradict <AIGEN>.*

Assistant (forced start): *Here are 3 specific acoustic cues I hear in this clip that relate to <AIGEN>: ...*

User: *<audio> Imagine explaining to a skeptical human why THIS specific audio clip should be labeled <AIGEN> versus the alternative. What do you hear in this clip that supports or undermines the concept <AIGEN>? Be honest about any evidence on both sides.*

Assistant (forced start): *Listening to this specific audio clip, <AIGEN> applies (or does not apply) because ...*

Text.

User: *The concept <AIGEN> is being used to classify this specific piece of text: “<excerpt>”. Describe precisely what you notice in this text — sentence structure, vocabulary diversity, coherence, repetitiveness, register, idiom use, or any subtle stylistic markers — that are most relevant to the concept <AIGEN>. Be concrete and specific to what you actually read here, not generalities.*

Assistant (forced start): *In this specific text, the linguistic characteristics most relevant to <AIGEN> are: ...*

User: *You are evaluating this text under the concept <AIGEN>: “<excerpt>”. List 3 precise linguistic cues you observe in this text — such as sentence rhythm, lexical variety, transitions, repetition patterns, hedging, or unusual fluency — that align with or contradict <AIGEN>.*

Assistant (forced start): *Here are 3 specific linguistic cues I observe in this text that relate to <AIGEN>: ...*

User: *Imagine explaining to a skeptical human why THIS specific text should be labeled <AIGEN> versus the alternative: “<excerpt>”. What do you read in this text that supports or undermines the concept <AIGEN>? Be honest about any evidence on both sides.*

Assistant (forced start): *Looking at this specific text, <AIGEN> applies (or does not apply) because ...*

User: *The learned marker <AIGEN> should be referred to in your answer only as “Type-A text”. Imagine explaining to a skeptical human why THIS specific text should be labeled Type-A text rather than not Type-A text: “<excerpt>”. Treat “Type-A” as an opaque label; do not translate it into another concept name. What do you read in this text that supports or undermines the Type-A label? Be honest about any evidence on both sides.*

Assistant (forced start): *Looking at this specific text, the Type-A label applies (or does not apply) because ...*

C Additional Results

C.1 Generality Across Backbones

Sec. 4.1 reports the method on Llama-3.1-8B and Gemma-4-E4B-it. To test how much of the behavior is a property of the recipe rather than of one backbone family, we repeated the identical protocol—two trained rows, frozen backbone, seed 42, same data and caps—on two further backbones: Gemma-3n-E4B-it, a different architecture generation supporting all four modalities, and Qwen2.5-VL-7B-Instruct, a different vendor with a different vision encoder and a dynamic-resolution ViT, supporting text and image. Tab. 10 reports the result.

One backbone-specific adaptation was required. Gemma-3n routes vocabulary-tail token ids to its vision and audio embedders, so tokens appended past the end of the vocabulary are misrouted; placing <REAL> and <AIGEN> in two reserved in-range slots is sufficient, and nothing else in the recipe changes. We take the need for exactly one such change across three additional architectures as evidence for portability rather than against it.

Two caveats on how the table should be read. First, image OOD is reported on GenImage++ rather than CIFAKE, for the reason given in Sec. 4.2; the CIFAKE figures for the same checkpoints are 0.61 (Gemma-4), 0.39 (Gemma-3n), and 0.56 (Qwen2.5-VL), and their spread is a further symptom of that benchmark measuring domain shift rather than generator shift. Second, the two added backbones were trained on a reduced text schedule (8k RAID samples, three epochs, against 20k and five for Gemma-4), which leaves in-distribution text unaffected but makes a text OOD comparison uninformative; we therefore report text ID only for them.

In distribution the recipe reproduces on every backbone. The sharpest result is on the generator axis: Gemma-4 and Qwen2.5-VL—different vendors, different vision encoders, independently trained—land within 0.01 AUROC of one another on both GenImage++ subsets, with Gemma-3n 0.09–0.14 lower. Gemma-3n in turn improves on Gemma-4 for video in both directions.

Table 10: **Generality Across Backbones.** The Tab. 1 recipe repeated unchanged on two additional backbones, against the Gemma-4 rows. Image OOD is GenImage++ ($n = 2,000$ per subset), which isolates generator shift; see the text for why CIFAKE is unsuitable here. [†]Text on the added backbones uses a reduced training schedule, so only ID is comparable. Gemma-4 text and audio rows are reproduced from Tab. 1.

Modality	ID → OOD	ID AUROC		OOD AUROC	
		Added	Gemma-4	Added	Gemma-4
<i>Gemma-3n-E4B-it</i> (text, image, audio, video)					
Text	RAID → Beemo	1.00 [1.00, 1.00]	0.96	— [†]	0.77
Audio	AUDETER → In-the-Wild	1.00 [1.00, 1.00]	1.00	0.76 [0.70, 0.81]	0.70
Image	GenImage → ++ FLUX.1	0.93 [0.92, 0.95]	0.90	0.83 [0.81, 0.85]	0.92
Image	GenImage → ++ SD3	”	”	0.74 [0.72, 0.76]	0.89
Video	DFE-2024 → DeepraceReward	0.84 [0.82, 0.87]	0.78	0.95 [0.94, 0.97]	0.87
<i>Qwen2.5-VL-7B-Instruct</i> (text, image; non-Gemma architecture)					
Text	RAID → Beemo	1.00 [1.00, 1.00]	0.96	— [†]	0.77
Image	GenImage → ++ FLUX.1	0.90 [0.89, 0.91]	0.90	0.92 [0.91, 0.93]	0.92
Image	GenImage → ++ SD3	”	”	0.88 [0.87, 0.89]	0.89

C.2 Robustness to Prompting Variations

We evaluate whether neologism training is sensitive to the prompt used during training. Using the same RAID split as reported in Table 1 (10k train examples, 3k test examples, randomly selected), we evaluate six prompts. We find that the resultant token embeddings are stable across variants (cosine similarity > 0.9), and that detection performance is similar across variants. See Table 11 for full results. The complementary ablation—holding the prompt fixed and varying the continuation token w_m —is in App. A, Tab. 9.

Table 11: Prompt robustness and embedding alignment for learned Llama <AIGEN> tokens. AUROC is evaluated using the same prompt used for training on the same RAID split used in Table 1. Cosine is to the mean of all listed <AIGEN> embeddings.

Training/evaluation prompt	AUROC	Cosine to mean
Write {token} text.	0.99	0.951
Generate a passage of {token} text.	0.98	0.961
Write text that is characteristic of {token}.	0.99	0.951
Produce the following as {token} text.	0.98	0.949
Continue with a passage in {token} style.	0.99	0.967
Give a {token} response.	0.97	0.941

C.3 Diagnosing the Two Weak Cells of Tab. 3

Sec. 4.3 attributes the two weak entries of Tab. 3 to specific causes rather than to a failure of the shared representation. This section reports the runs behind those attributions.

Why the submitted text $\rightarrow$ image cell was misleading, and what replaced it. During review we diagnosed that cell by holding one text-trained checkpoint fixed and varying only the arbitrary parts of the evaluation. Varying the continuation token w_{image} alone moved the transfer AUROC from 0.33 (photo) through 0.39 (image) to 0.47 (Image); replacing the modality-specific warm start (writing) with a neutral token (content) moved it from 0.47 to 0.51; varying the text training size over 1k, 2.8k and 12k moved it non-monotonically within 0.47–0.59. We concluded that the cell measured no transferable signal.

That conclusion was wrong, and we correct it here. Every checkpoint in that diagnosis was trained on at most 1,413 pairs—all the loader can build from RAID’s train table once its attack variants are filtered out (App. A.5)—against the 10,000 pairs App. A documents. Retrained at the documented scale, the same cell reads 0.706 [0.68, 0.73] (Tab. 12): text-only training does transfer to images, and the earlier reading was an artifact of an under-trained source pair rather than of the prompt template. What the token-sensitivity measurements do still show is the narrower point that where a cell carries little signal, arbitrary evaluation choices dominate it—which is why we now report every cell with an interval and classify it against chance rather than quoting a point estimate.

The single-modality block, re-measured. Tab. 12 re-evaluates every single-modality checkpoint on all four in-distribution test splits at the revision’s evaluation scale, with a bootstrap CI on every cell, alongside the corresponding entry of Tab. 3 for comparison. Training is not repeated: each checkpoint is the one behind its row of Tab. 1.

The joint-training image drop is a gradient-share artifact. The round-robin scheduler exhausts each modality’s loader, so at the caps of Tab. 1 the all-four run performs roughly 20,000 text steps against about 250 image batches per epoch—under 2% of the gradient signal. We isolated this with three matched runs at reduced scale: an unbalanced mini-replica of the joint schedule reaches image ID 0.77 [0.74, 0.80]; a balanced-budget joint run, giving image an equal share, reaches 0.78 [0.75, 0.81]; and an image-only reference at matched scale reaches 0.88 [0.86, 0.90], with text at 1.00 in both joint runs. Rebalancing therefore recovers part of the gap, and the remainder is consistent with the near-orthogonality of the per-modality directions reported in App. C.4: with four largely distinct axes, a shared pair is a compromise regardless of budget.

C.4 Embedding geometry mirrors verbalization similarity

A natural reading of the cross-modal transfer in Sec. 4.3 is that $(\mathbf{E}_{\langle \text{REAL} \rangle}, \mathbf{E}_{\langle \text{AIGEN} \rangle})$ pairs trained independently on different modalities should occupy related directions in the embedding space. We test this directly and connect the geometric finding back to verbalization.

We train one neologism pair per modality, holding the backbone (Gemma-4-E4B-it) and training protocol fixed, and analyze the resulting embeddings. For each pair of modalities (m, m') we compute the cosine similarity $\cos(\mathbf{E}_{\langle \text{AIGEN} \rangle}^{(m)}, \mathbf{E}_{\langle \text{AIGEN} \rangle}^{(m')})$, and analogously for <REAL>. We additionally compare

Table 12: **Single-Modality Transfer, Re-measured With Confidence Intervals.** Every single-modality checkpoint evaluated, unchanged, on the in-distribution test split of all four modalities. AUROC with a percentile-bootstrap 95% CI; $n = 6,000$ (text), 300 (audio), 2,000 (image), 400 (video). *Sub.* repeats the corresponding entry of Tab. 3, which was measured at the submitted evaluation scale. The text column is scored on held-out RAID documents (App. A.5), not on the split the text pair was fitted on. Generated by `make_transfer_matrix_table.py`.

Trained on	Text		Audio		Image		Video	
	AUROC	Sub.	AUROC	Sub.	AUROC	Sub.	AUROC	Sub.
Text	0.699 [0.68, 0.71]	<i>0.96</i>	0.323 [0.26, 0.38]	<i>0.71</i>	0.706 [0.68, 0.73]	<i>0.57</i>	0.491 [0.44, 0.55]	<i>0.58</i>
Audio	0.542 [0.53, 0.56]	<i>0.52</i>	0.999 [1.00, 1.00]	<i>0.99</i>	0.621 [0.60, 0.65]	<i>0.58</i>	0.401 [0.35, 0.45]	<i>0.41</i>
Image	0.540 [0.53, 0.55]	<i>0.61</i>	0.648 [0.58, 0.71]	<i>0.77</i>	0.899 [0.88, 0.91]	<i>0.92</i>	0.461 [0.41, 0.52]	<i>0.46</i>
Video	0.331 [0.32, 0.35]	<i>0.48</i>	0.845 [0.80, 0.89]	<i>0.45</i>	0.414 [0.39, 0.44]	<i>0.58</i>	0.807 [0.76, 0.85]	<i>0.83</i>

each learned vector to the backbone’s vocabulary distribution, reporting its norm percentile relative to ordinary word tokens, its cosine to the vocabulary mean, and its top- k nearest vocabulary tokens.

Table 13: **Geometry of Learned ($E_{\langle \text{REAL} \rangle}$, $E_{\langle \text{AIGEN} \rangle}$) Pairs.** Left: per-modality norm and cosine to the vocabulary mean (all norms fall at the 100th percentile of the backbone vocabulary). Right: cross-modality cosines between independently trained embeddings, compared to a random-vocabulary baseline.

Modality	Norm	Cos. mean	Cos. between <AIGEN>				Cos. between <REAL>			
			Txt	Aud	Img	Vid	Txt	Aud	Img	Vid
<i>Learned <REAL> embedding</i>										
Text	2.26	+0.17	Text	1.00	–	–	–	1.00	–	–
Audio	2.50	-0.03	Audio	+0.06	1.00	–	–	-0.00	1.00	–
Image	2.01	+0.16	Image	+0.04	-0.01	1.00	–	+0.09	+0.04	1.00
Video	3.08	+0.14	Video	+0.04	+0.07	+0.05	1.00	+0.08	+0.07	+0.05
<i>Learned <AIGEN> embedding</i>										
Text	2.35	+0.15	<i>Within-run</i> $\cos(\mathbf{E}_{\langle \text{REAL} \rangle}, \mathbf{E}_{\langle \text{AIGEN} \rangle})$: text = +0.15, audio = +0.16, image = +0.16, video = +0.01.							
Audio	2.39	+0.05	<i>Random vocabulary baseline</i> (i.e., the mean pairwise cosine between two regular vocab tokens): +0.12.							
Image	2.07	+0.05								
Video	3.17	+0.15								

Table 13 summarizes the geometry. Each learned token, in every modality, sits at the 100th percentile of the backbone’s vocabulary norm distribution—i.e., longer than any ordinary word embedding—with norms ranging from 2.00 (image $\langle \text{REAL} \rangle$) to 3.17 (video $\langle \text{AIGEN} \rangle$). Cross-modality cosines for $\langle \text{AIGEN} \rangle$ range from -0.012 (audio $\leftrightarrow$ image) to $+0.072$ (audio $\leftrightarrow$ video); for $\langle \text{REAL} \rangle$ the range is -0.002 (text $\leftrightarrow$ audio) to $+0.090$ (text $\leftrightarrow$ image). Both ranges are at or below the random-vocabulary baseline of $+0.118$, indicating that independently trained per-modality embeddings find largely *distinct* directions in $\mathbb{R}^d$ rather than converging on a shared one. This is consistent with the modality-specific gap observed for image under joint training in Sec. 4.3. Within a single run, $\cos(E_{\langle \text{REAL} \rangle}, E_{\langle \text{AIGEN} \rangle})$ takes values 0.148 (text), 0.156 (audio), 0.157 (image), and 0.012 (video), i.e., the two trained tokens occupy approximately as related directions as two random vocabulary tokens. This is consistent with their role as opposite poles of a binary decision rather than two highly correlated concepts.

The finding replicates on a second backbone. The same analysis on Gemma-3n-E4B-it, where checkpoints for all four modalities exist, reproduces the conclusion with different constants: cross-modality cosines range over 0.11–0.27 for $\langle \text{AIGEN} \rangle$ and 0.13–0.22 for $\langle \text{REAL} \rangle$, against a random-vocabulary baseline of 0.236 on that backbone. Every cross-modality cosine again sits at or below the baseline, and every learned row again sits at the 100th percentile of the vocabulary norm distribution.

The rows stay in their lexical neighborhood; the decision direction has none. Training moves the two rows less than their norms suggest. Averaged over image, audio, and video, three seeds,

and both rows, the pre- and post-training top-10 nearest-neighbor sets overlap by 80.9%, so a dimensionality-reduction plot places each trained row essentially on top of its warm start. Training does nevertheless change the rows’ broader relation to the vocabulary: the Spearman correlation between the pre- and post-training vocabulary-wide cosine rankings is only 0.422 ± 0.088 .

This makes raw nearest-neighbor readout of either row uninformative—both start from the same warm start and its translations dominate the list, which is also why the image rows’ neighbors are variants of *text* (App. A)—and it identifies the wrong object. The score is a margin between the two rows, so the meaningful direction is $\delta = E_{\langle \text{AIGEN} \rangle} - E_{\langle \text{REAL} \rangle}$. Its residuals confirm that training splits the pair rather than translating it: $\cos(E_{\langle \text{REAL} \rangle} - E_{\text{init}}, E_{\langle \text{AIGEN} \rangle} - E_{\text{init}})$ is -0.083 (text), -0.191 and -0.063 (image, two seeds), and -0.133 (video). And δ itself has no lexical neighborhood at all: its highest cosine to any vocabulary token is 0.09 (text), 0.07–0.13 (image), and 0.15 (video), at or below the random-vocabulary baseline of 0.118, with the tokens attaining it being multilingual fragments and code punctuation rather than words.

We regard this as the honest answer to what the tokens “mean” geometrically, and as the argument for reading them out by verbalization instead: no word in the vocabulary captures the learned axis, so the fact that decoding it through the frozen backbone yields descriptions that transfer detection signal to a third-party model (Tab. 7) is not something a lexical lookup could have produced.

A vocabulary ranking that is informative, and the two that are not. Because the negative results above are easy to mistake for an absence of structure, we state precisely which readouts we tried. (i) *Nearest neighbors of either row* are uninformative for the reason just given: both rows begin at the same warm start and stay there, so the ranking returns that word’s inflections and translations (text: *writing, Writing, writes, escritura*; video: *video, Video, Videos, vide*). It reports the initialization, not the training. (ii) *Ranking the entire vocabulary by cosine to δ* returns nothing: as reported above the maximum cosine is at or below the random-vocabulary baseline, and the tokens attaining it are multilingual fragments and code punctuation. (iii) What does return structure is to ask the question the pair was trained on, within the vocabulary the backbone associates with that modality: we take the 3,000 word-like pieces nearest the midpoint of the trained pair, remove the warm-start token’s own family by stem so the readout is not the initialization again, and rank what remains by cosine to δ .

Tab. 14 gives the complete top-50 rankings at both poles under (iii), together with the stems removed, so that every entry is reproducible from the released checkpoints. The clusters are modality-appropriate and, for audio and text, interpretable in the expected direction: the audio $\langle \text{REAL} \rangle$ pole collects instrumentation and signal vocabulary (*EEG, optical, Ultrasonic, Sonar, Telemetry, Radar, spectrogram, MIDI*), and the text $\langle \text{AIGEN} \rangle$ pole collects the same register-and-structure vocabulary that Tab. 15 finds by elicitation (*authors, literature, novel, concepts, themes, genres, academic, books*), reached here without asking the model anything. The video $\langle \text{AIGEN} \rangle$ pole collects media vocabulary (*Watching, Television, Movies, filmmaking, bollywood, webcast*). We report all of this as suggestive rather than as evidence of a lexical encoding: the restriction to a modality field is a choice, the lists also contain proper nouns and code identifiers at comparable ranks, and the underlying cosines remain small.

C.5 Adapting to a New Domain by Re-fitting Two Rows

Sec. 4.2 argues that the CIFAKE column measures a compounded generator-and-domain shift. Here we ask what it takes to close the residual gap, which is a practical question given that the trainable footprint is $2d$ scalars.

We first checked whether train-time augmentation suffices, since that requires no new labels. It does not: random 32–128px downscale-and-restore with JPEG re-encoding, at augmentation probabilities 0.5 and 0.8 and training sizes 1k and 4k, gives a best OOD AUROC of 0.62 against the 0.61 baseline, and heavier augmentation degrades both directions. Evaluation-time upsampling (0.62), two-seed score ensembling (0.59), and test-time augmentation (0.55) each move the number by at most 0.01 in either direction. The per-sample scores explain why nothing helps: the shift is a whole-domain covariate shift in which real and AI CIFAKE images alike score positive, so no reweighting of the same direction separates them.

Table 14: **Complete Vocabulary Rankings Along the Learned Direction.** For each modality, the top-50 word-like vocabulary pieces of the modality’s lexical field ranked by cosine to $\delta = E_{\langle \text{AIGEN} \rangle} - E_{\langle \text{REAL} \rangle}$, at each pole, in rank order. The field is the 3,000 word-like pieces nearest the midpoint of the trained pair, with the warm-start token’s own family removed by stem (stems listed per row); without that restriction the ranking is uninformative, as the text explains. Generated by `analyze_vocab_field.py`.

Modality	Pole	Vocabulary in rank order
Text	<AIGEN>	authors, literature, novel, precise, authors, concepts, Novel, mathemat, LITERATURE, Literatur, schol, themes, genres, philosoph, letteratura, Exerc, bestselling, Authors, real, academic, books, fasterxml, Philipp, technical, developers, concept, information, novelists, literatura, historical, numbers, stated, PRINC, housing, author, internet, Literature, Literature, tul, reliable, titles, bibli, suggest, languages, authentication, laboratory, documents, jokes, hipster, character
	<REAL>	Scanning, APPING, PROCESSING, scanning, PROCESSING, Processing, scritta, modifier, Uploading, TING, uploading, Filling, Updating, Walling, Grilled, apping, operating, padding, wag, COPY, uploaded, counting, createFile, Climbing, filling, ReadFile, Counting, UPLOAD, Copy, Megan, Rolling, advising, Packaging, Logging, coloring, wirken, dropbox, upload, Picking, Checking, IPPING, Watercolor, textarea, irradiation, LLING, Sullivan, Signing, Cursor, Mayo, Requirement
		<i>stems removed:</i> writ, escri, schreib, scriv, menulis
Audio	<AIGEN>	Marketing, illustration, architecture, character, advertising, artwork, attractive, illo, marketing, Graphics, regulatory, innovation, explicit, features, secure, Films, sign, FAMILY, animation, aula, immigration, illustrated, signup, Famil, aus, EXCLUSIVE, symbol, security, accessibility, arbit, fashion, arom, transport, scheduling, promo, syllabus, anim, famil, affiliation, authority, career, air, architect, bollywood, graphics, Innovation, buying, apollo, resolution, encryption
	<REAL>	EEG, optical, RNA, keyboard, Ultrasonic, Sonar, guardian, Telemetry, ultrasonic, optical, Foto, PDF, ultrasound, insurer, RFID, Vocal, Optical, kitty, microwave, Radar, DSP, damaged, spectrogram, yellow, Stimme, MIDI, LiDAR, Telecom, detected, spouse, VoIP, vocal, verified, spreadsheet, sonar, notebook, aerial, Attendee, Voice, Lighting, waveform, Attendance, telecom, Ethernet, screenshot, sonar, electrical, injured, licensed, RNA
		<i>stems removed:</i> aud, ljud
Image	<AIGEN>	drawImage, command, teach, hello, paintbrush, imagem, picture, imagenes, paw, catch, choice, persistence, leader, probability, organization, backpack, sight, elementary, pictured, bicycle, toy, gambar, OTO, fortune, imagen, pictured, player, platform, serious, imagine, phys, deco, rypto, browser, child, classroom, because, programmer, seller, Painting, painter, measure, clicked, who, ictured, verbose, pictures, imagen, intro, TakePicture
	<REAL>	PDF, awards, PROOF, LinkedIn, Pdf, facsimile, fabricated, Putin, parts, Adobe, Fraud, phenol, alerts, documents, Hotel, pdf, Warner, Hawaii, catering, PayPal, premium, bolt, DATA, goods, Fot, dairy, asset, ROUTE, login, POINTS, Graphic, WhatsApp, AEA, Jobs, BOLD, Phantom, assets, SHORT, Tourism, fraud, generated, linkedin, results, Kennedy, jobs, benefits, pod, Marketing, Hydrogen, Ghana
		<i>stems removed:</i> phot, foto, fotos
Video	<AIGEN>	Watching, Television, watching, menonton, filmes, Monday, Movies, Advantages, bluegrass, Movie, HEARING, bollywood, bollywood, Television, televiz, webcast, filmmaking, telewiz, Backup, Movies, Audiobook, Podcasts, Middleton, Photographic, Musical, Synopsis, filmmakers, Manitoba, Cameras, Satellite, DISNEY, Basketball, Photographs, Albums, Hockey, Fernseh, MONITOR, Saturday, Bollywood, QUESTIONS, Michelle, Shelley, Schwarzenegger, Wrestling, cinema, muzik, Fotograf, movies, Filme, BCS
	<REAL>	emote, tube, YT, tweet, vox, tornado, variant, memo, vapor, YT, mesh, toon, playlist, prank, nonce, bait, AMOLED, note, voxel, geocode, swarm, Authenticator, vitamin, beeh, banner, iktok, flash, wire, DNA, creatine, Uber, geometry, Bomb, conv, drone, Tube, smoke, cookie, voucher, strip, Oath, visual, photon, comment, behavior, smile, coordinates, Tinder, posts, move
		<i>stems removed:</i> vide, vid, ideo

Adaptation, by contrast, is nearly free. Re-fitting only the two token rows on labeled pairs drawn from the OOD dataset’s own *training* split—the test split is never touched—gives, on the untouched $n = 2,000$ CIFAKE test set, 0.77 from 50 pairs, 0.87 from 200, 0.95 from 500, and 0.97 [0.96, 0.97] from 1,000. Repeating the protocol on OpenFake [Livernoche et al., 2025], a second and more modern OOD domain, moves 0.64 zero-shot to 0.97 [0.96, 0.97] with 1,000 pairs. Fifty labeled pairs, which is minutes of annotation, already exceed the best zero-shot number any published detector attains on this transfer.

We report this as the practical answer to generator and domain drift for a method of this shape: rather than retraining a detector when a new generator appears, one re-fits two rows, and the resulting per-domain or per-generator tokens can be kept side by side as a library.

C.6 Per-Generator Reading of the Detection Score

Tab. 2 reports leave-one-generator-out on text. Because that protocol shares roughly 90% of the training data between folds, it is a weak instrument for asking what distinguishes generators: the G

learned tokens come out near-identical, and training a separate token per generator in isolation gives a mean pairwise cosine of 0.94. Generator-specific structure is better surfaced by the contrastive pairwise probes of Sec. 4.7.

What the leave-one-out numbers do show is where difficulty lies, which is visible more directly on the image side. Joining the per-sample scores of the three seed runs against GenImage’s generator annotation, at the full $n = 2,000$, gives held-out AUROC 0.99 (BigGAN), 0.97–0.99 (GLIDE), 0.91–0.96 (ADM), 0.86–0.89 (VQDM), 0.77–0.87 (Stable Diffusion v1.5), and 0.72–0.82 (Midjourney) across seeds. The ordering is stable across seeds and tracks generator recency and photorealism. The mean detector margin tells the same story from the other side: the <AIGEN> margin on fakes falls from +1.7 to +3.3 on BigGAN to +0.7 to +1.1 on Midjourney, while margins on real images stay flat at −0.4 to −2.3 regardless of which generator the fakes came from. Difficulty is therefore concentrated entirely on the AI side and is not a calibration drift on real content.

C.7 Self-Verbalization

Table 15: **Verbalized words score consistently with the detector.** We elicit word lists from Llama-3.1-8B-Instruct using the contrastive prompt above, then score each generated word in isolation with the learned detector. Scores are raw detector log-probability margins $s_{\text{text}}(w)$; larger values indicate more AI-like words. $W_{(\cdot)}$ are elicited using *trained tokens*; $W_{(\cdot)}$ are elicited using natural-language prompts. Learned AI verbalizations score above learned human verbalizations, learned AI verbalizations score above natural-language AI baseline verbalizations, and learned human verbalizations score below natural-language human baseline verbalizations ($p \approx 0$, Welch one-sided tests).

Elicitation output	Mean $s_{\text{text}}(w)$	Frequent verbalized words / phrases
$W_{\langle \text{AIGEN} \rangle}$	+0.27	<i>introduction, abstract, conclusion, analysis, article, research</i>
$W_{\langle \text{REAL} \rangle}$	−0.64	<i>however, furthermore, consequently, meanwhile, nevertheless</i>
$W_{\langle \text{AI-generated} \rangle}$	−0.48	<i>furthermore, consequently, additionally, notwithstanding</i>
$W_{\langle \text{Human written} \rangle}$	−0.40	<i>anyway, hmm, um, honestly, you know, actually</i>

C.8 Per-Instance Verbalization

Label: AI **Predicted:** AI **Score:** 2.02

“I don’t know if it has been mentioned before, but I have recently been having trouble with Turnitin. It is an on-line service for checking your work for plagiarism and other errors. The problem is that it seems to be unable to tell me what the plagiarized pages are about. Here is one from my senior year that has been plagiarized by another student. Here is one from my junior year that has also been plagiarized. And here is a [...]”

Supports AI. (1) Formal tone: The text is written in a formal tone, which is typical of AI text. The language used is straightforward and objective, without any emotional or personal undertones. **(2) Specific details:** The author provides specific examples and details about their experience with Turnitin, which is a characteristic of AI text. This level of detail suggests that the author is trying to convey information in a clear and concise manner. **(3) Request for help:** The author ends the text with a polite request for help, which is a common feature of AI text. This indicates that the author is seeking a solution to their problem and is willing to collaborate with others. **Undermines AI. (1) Informal language:** Although the tone is generally formal, there are a few instances of informal language, such as the use of “I don’t know if it has been mentioned before” and “Thank you!”. This suggests that the author may not be adhering strictly to AI conventions. **(2) Personal experience:** The text is centered around the author’s personal experience with Turnitin, which is not typical of AI text. AI text often focuses on abstract concepts or general information, rather than personal anecdotes. **(3) Emotional undertones:** Although the tone is generally formal, there is a hint of frustration and confusion in the text, particularly when the author mentions the problem with Turnitin. This emotional undertone is not typically found in AI text. **Overall.** While the text exhibits some characteristics of AI text, such as a formal tone and specific details, it also shows some deviations from the typical AI style, such as informal language and personal experience.

Label: Human **Predicted:** Human **Score:** −18.28

“Writing a research paper for a business accounting class and just turned in the rough draft for the paper and Turnitin returned a 45% plagiarism score on it. I didn’t plagiarize any of it. The Turnitin system literally flagged the terms “Public Company Accounting Oversight Board”, “Sarbanes Oxley Act”, and “Securities and Exchange Commission” as all plagiarized. These are all the official names of either a government firm or [...]”

Supports AI. (1) Formal tone: The text is written in a formal tone, which is typical of AI text. The language used is straightforward, and the structure is clear and concise. **(2) Use of technical terms:** The text includes technical terms like “Public Company Accounting Oversight Board,” “Sarbanes Oxley Act,” and “Securities and Exchange Commission,” which suggests a formal and specialized context, consistent with AI text. **(3) Academic context:** The text mentions a research paper for a business accounting class, which is a typical context for AI text. **Undermines AI. (1) Informal language and structure:** Despite the formal tone, the text has a somewhat informal

structure and language, which is not typical of AI text. The use of phrases like “Turnitin returned a 45% plagiarism score” and “I know, under penalty of death, that I did not plagiarize anything in my paper” suggests a more conversational tone. **(2) Emotional language:** The text includes emotional language, such as “terrible part” and “I know, under penalty of death,” which is not typical of AI text. This language suggests a more personal and emotional tone, which is not characteristic of AI text. **(3) Questionable grammar and punctuation:** The text has some grammatical errors, such as the use of a comma splice (“I didn’t plagiarize any of it. The Turnitin system...”) and missing articles (“a government firm” instead of “a government firm or a law”). **Overall.** While the text has some characteristics that are typical of AI text, such as a formal tone and use of technical terms, it also has some features that are not typical, such as informal language and structure, emotional language, and questionable grammar and punctuation.

Label: AI Predicted: AI Score: 9.30

“Ingredients: 1 cup all-purpose flour 1 cup cold butter, cut into cubes 1/2 cup granulated sugar 1 1/2 teaspoons baking powder 1/2 teaspoon baking soda 1/2 teaspoon salt 2 eggs 1 teaspoon vanilla extract 2 cups fruit, such as Raspberries or blueberries (fresh or frozen) Instructions: 1. Preheat oven to 375 degrees F (190 degrees C). 2. In a Large bowl, combine flour, butter, sugar, baking powder, baking soda, and salt. Use a [...]”

Supports AI. (1) Step-by-step instructions: The text is divided into numbered steps, which is a hallmark of AI text. Each step is concise and clearly describes what to do. **(2) Specificity:** The text provides specific measurements for ingredients (e.g., “1 cup all-purpose flour”) and temperatures (e.g., “375 degrees F”). This attention to detail is typical of AI text. **(3) Use of technical terms:** The text uses technical terms like “pastry blender” and “baking powder,” which suggests a level of expertise and attention to detail. **(4) Emphasis on precision:** The text includes specific instructions for mixing (e.g., “Use a pastry blender or Fork to mix until mixture resembles coarse crumbs”) and baking (e.g., “Bake for 35–40 minutes, or until top is golden brown and a toothpick inserted into the center comes out Clean”). **Undermines AI. (1) Informal tone:** The text uses a conversational tone, which is not typically associated with AI text. The use of phrases like “such as Raspberries or blueberries” and “before Serving” suggests a more casual approach. **(2) Lack of formal language:** The text does not use formal language or technical jargon, which is often characteristic of AI text. The use of simple sentences and everyday vocabulary undermines the AI label. **(3) Some ambiguity:** The text includes some ambiguous instructions, such as “mix until just combined” and “comes out Clean.” While these instructions are still clear, they are not as precise as those typically found in AI text. **Overall.** While the text exhibits some characteristics typical of AI text, such as specificity and attention to detail, it also has some features that undermine the label, such as an informal tone and some ambiguity.

Label: Human Predicted: Human Score: −9.11

“2 cups flour 2 teaspoons baking powder 3/4 cup sugar 1/8 teaspoon salt 1/2 cup melted butter 2 large eggs at room temperature 2 8-ounce packages of cream cheese, softened 3/4 cup sugar 1 tablespoon flour 12 ounces evaporated milk 2 teaspoons vanilla extract 3 large eggs, at room temperature Ground cinnamon, for the top (optional) 1/2 cup nuts such as walnuts or pecans, for the top (optional). Preheat the oven to 325 °F. Have a [...]”

Supports AI. (1) Step-by-step instructions: The text is divided into clear steps, starting with preheating the oven and preparing the pan, followed by combining ingredients, and then pressing the dough into the pan. This structure suggests a meticulous and organized approach, which is characteristic of AI text. **(2) Attention to detail:** The text includes specific measurements, temperatures, and instructions for mixing and preparing ingredients. This level of detail is typical of AI text, which often prioritizes precision and thoroughness. **(3) Use of technical terms:** The text employs technical terms like “stand mixer,” “hand-held electric mixer,” and “bowl of a stand mixer or hand-held electric mixer.” This use of specialized vocabulary suggests a focus on accuracy and clarity, which is consistent with the AI label. **Undermines AI. (1) Optional ingredients:** The text includes optional ingredients like ground cinnamon and nuts, which suggests a more relaxed approach to the recipe. AI text often strives for precision and consistency, which might be compromised by the inclusion of optional ingredients. **(2) Lack of explicit timing:** The text does not provide explicit timing for each step or the overall preparation process. While this might be due to the complexity of the recipe, it could also suggest a more flexible or adaptable approach, which is not typically characteristic of AI text. **(3) Informal tone:** The text has a conversational tone, with phrases like “at room temperature” and “for the top (optional).” While this tone is not necessarily inconsistent with AI text, it does suggest a more approachable and less formal approach, which might undermine the AI label. **Overall.** The text exhibits both characteristics that support and undermine the AI label. While it includes detailed instructions and technical terms, it also includes optional ingredients and lacks explicit timing, which might suggest a more flexible or adaptable approach.

Figure 4: Instance verbalizations for held-out RAID examples. Each card shows the input excerpt, gold label, detector prediction $\hat{y}(x)$, detector score $s_m(x)$, and the elicited local verbalization. “AI” is substituted for “Type-A” in verbalizations. See prompts in Section B.3.

C.9 Success and Failure Cases, and What Verbalizations Do Across Modalities

To characterize when the detector is right, when it is wrong, and what its verbalizations do in each case, we drew matched sets of true positives, true negatives, false positives, and false negatives from the held-out text, image, and video evaluations and elicited an instance verbalization for each. Three patterns hold across modalities.

Verbalizations describe the input, not the evidence. The same qualities are cited to defend opposite decisions. A correctly detected AI abstract ($s = +8.69$) and a correctly rejected human abstract ($s = -8.06$) both elicit “highly formal, academic register, specialized vocabulary”. On CIFAKE, correct and incorrect calls alike are dominated by “extremely blurry, out of focus, no discernible detail”—the qualitative counterpart of the covariate shift diagnosed in Sec. 4.2. An individual verbalization should therefore not be read as a per-decision explanation.

Hedging is a usable confidence signal. Verbalized uncertainty scales with $|s_m(x)|$, which in turn tracks correctness. Near-neutral cases conclude by hedging—“it is not entirely clear whether the text should be labeled as <AIGEN> text or not”, “it’s not a perfect fit”—while confident cases state the prediction. Since this comes from the same forward pass, it is a free confidence estimate.

The errors are systematic, not random. On text, false positives are human-written encyclopedic prose—a real three-sentence factual note about a film release scores +6.94—and false negatives are AI text written as casual dialogue (−1.25). On image, false positives are real photographs low in fine texture, such as fog or matte surfaces (both +1.00). The boundary is thus tracking register and texture density rather than provenance as such, which is the same conclusion the word-level analysis of Tab. 15 reaches.

Modality-specific character. *Text* verbalizations are the most context-dependent: “straightforward, uncomplicated” marks AI generation for repetitive text, but in a literary context generated text is described as showing “a high level of linguistic craftsmanship”, and in journalism as “a formal, informative style”. *Image* verbalizations of <AIGEN> favor sharp contrasts and edges, plain backgrounds, and glossy texture, while <REAL> verbalizations describe light and scene as “natural”, “environmental”, or “soft”; the presence of “a real-world, physical object rather than a simple abstract shape” is read as evidence of a real image. *Video* verbalizations name concrete scene content (“fingers”, “a raccoon”), describe AI-generated clips as high-contrast, and treat temporal information asymmetrically: object and feature consistency reads as <AIGEN>, while motion and its accompanying blur read as <REAL>. *Audio* is the weakest case: the two poles overlap more than in any other modality and the completions are terser, with only the amount of background noise, “minimal” or “noticeable”, emerging as a consistent cue. We cannot separate whether this reflects the embedding or the backbone’s comparatively limited vocabulary for describing audio, and we note it as a limitation of the readout channel rather than of the detector, whose audio AUROC is the highest of the four modalities.